Reading Supplement
Module-II Biomolecules

Biomolecules

1. Composition of Living Matter
2. Water
3. Carbohydrates
4. Lipids
5. Proteins
6. Nucleic Acid
7. Vitamins and Minerals

---

1. COMPOSITION OF LIVING MATTER

- All matter on earth is known to composed of basic chemical elements i.e., atoms. Living matter is essentially composed of carbon, hydrogen and oxygen. In other words, all living organisms, from unicellular bacteria to multicellular humans, are made up of chemical substances of both organic and inorganic types.

- With the exception of water, most of the molecules in living organisms contain the element carbon. These complex carbon-containing molecules are known as **organic** compounds. In the majority of cases, carbon (C) is joined with hydrogen (H), oxygen (O), and in some cases, nitrogen (N) and phosphorus (P). The organic compounds most commonly found in cells are carbohydrates, lipids, proteins, and nucleic acids.

- Other compounds are classified as inorganic. Of the inorganic compounds, water and minerals are among the most important. Mineral exists as salts of inorganic compounds. It contains cations (other than H^+) and anions (other than OH^-). Salts are electrolytes which conducts electrical current. These salts can dissociate in water into their components. Carbon dioxide is termed 'inorganic' as it is not a complex molecule and does not contain hydrogen. Living things contain both organic and inorganic compounds.

- As of July 2013, 118 chemical elements have been identified, the latest being ununseptium in 2010. Of the 118 known elements, only the first 98 are known to occur naturally on Earth. Out of known 98 naturally occurring elements only about 30 elements are essential to living organisms.

- The six most abundant elements in living organisms, in terms of the percentage of the total number of atoms, **are carbon, hydrogen, oxygen, nitrogen, phosphorous and sulphur**, which together make up over 99% of the mass of most cells.

- They are the lightest elements capable of forming one, two, three, and four bonds, respectively. In general, the lightest elements form the strongest bonds.

NATURALLY OCCURRING ELEMENTS IN THE HUMAN BODY

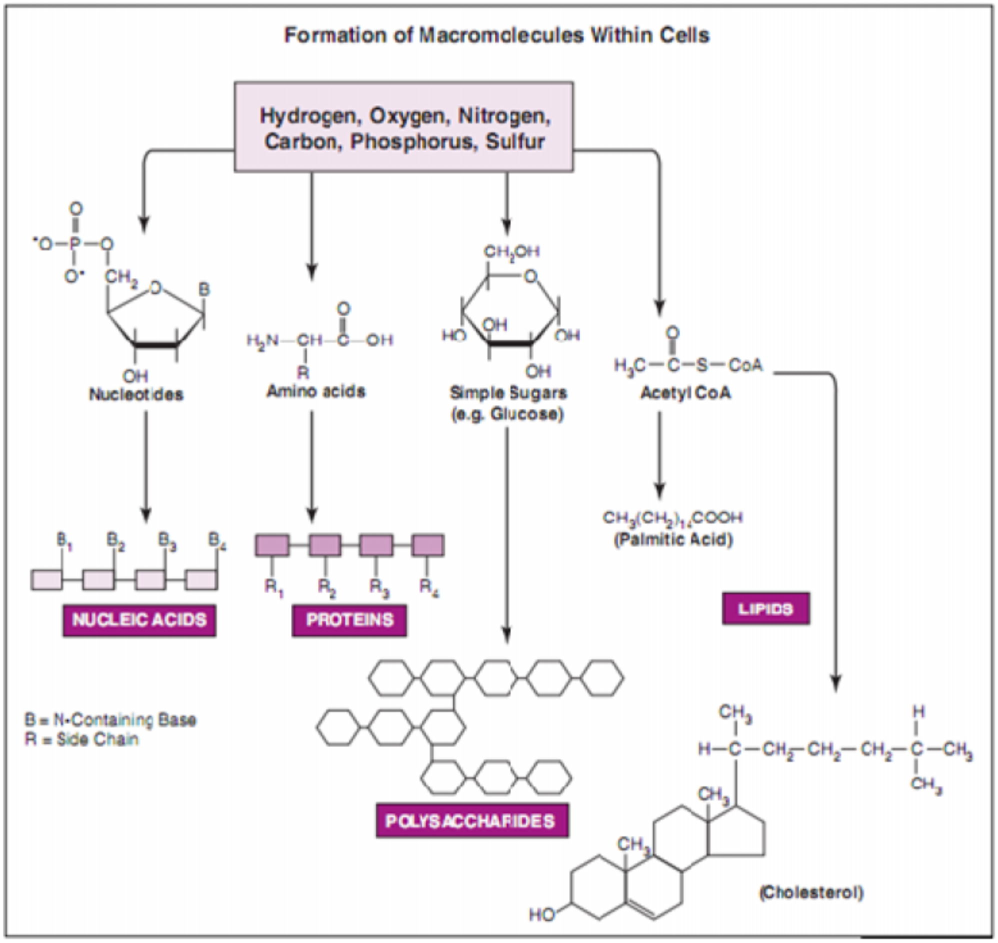
Symbol	Element	Wet Weight Percentage*	
O	Oxygen	65.0	96.3
C	Carbon	18.5	
H	Hydrogen	9.5	
N	Nitrogen	3.3	
Ca	Calcium	1.5	
P	Phosphorus	1.0	
K	Potassium	0.4	
S	Sulfur	0.3	
Na	Sodium	0.2	
Cl	Chlorine	0.2	
Mg	Magnesium	0.1	

Trace elements (less than 0.01%): boron (B), chromium (Cr), cobalt (Co), copper (Cu), fluorine (F), iodine (I), iron (Fe), manganese (Mn), molybdenum (Mo), selenium (Se), silicon (Si), tin (Sn), vanadium (V), and zinc (Zn).

*Includes water.

Table 1.1 Percentage compositions of chemical elements in living matter

- Covalently linked carbon atoms in biomolecules can form linear chains, branched chains, and cyclic structures. To these carbon skeletons are added groups of other atoms, called functional groups, which confer specific chemical properties on the molecule.
- Molecules with covalently bonded carbon backbones (organic compounds) occur in limitless variety. Since most biomolecules are organic in nature; we can therefore infer that the bonding versatility of carbon was a major factor in the selection of carbon compounds for the molecular machinery of cells during the origin and evolution of living organisms.
- No other chemical element can form molecules of such widely different sizes and shapes or with such a variety of functional groups.
- When these six major elements combined in various ways, form virtually all known organic biomolecules. They are initially utilized in the synthesis of a small number of building blocks that are, in turn, used in the construction of a vast array of vital macromolecules.
- There are four general classes of macromolecules within living cells: nucleic acids, proteins, polysaccharides, and lipids.
- These compounds, which have molecular weights ranging from 1 x 10^3 to 1 x 10^6, are created through polymerization of building blocks that have molecular weights in the range of 50 to 150.
- Although subtle differences do exist between cells (e.g., erythrocyte, liver, muscle or fat cell), they all generally contain a greater variety of proteins than any other type of macromolecule, with about 50% of the solid matter of the cell being protein (15% on a wet-weight basis).



- Cells generally contain many more protein molecules than DNA molecules, yet DNA is typically the largest biomolecule in the cell. About 99% of cellular molecules are water molecules, with water normally accounting for approximately 70-80% of the total wet-weight of the cell.
- Although water is obviously important to the vitality of all living cells, the bulk of our attention is usually focused on the other 1% of biomolecules.

2. WATER

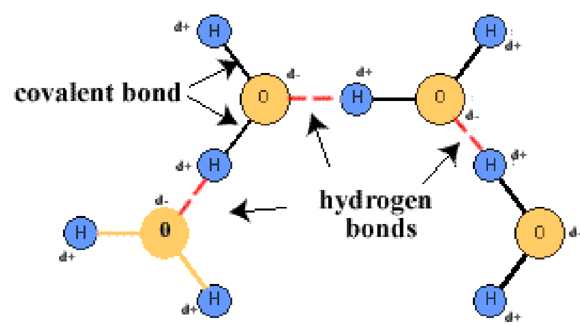
- Of the smaller molecules, water is the most abundant, typically making up between 60-95% of the fresh mass of an organism.
- They are the kind of molecules which biologist speculate could have been made in the “primeval soup” of chemicals which is thought to have existed in the early history of the planet, before life itself appeared.
- Water is important for two reasons;
  (a) It is a vital chemical constituent of living cells;
  (b) It provides an environment for those organisms that live in water.
- Among various important properties of water, Polarity and Hydrogen bonding between its molecules, seems to be the most fundamental comparatively.

2.1 Properties of Water

Polarity and Hydrogen bonding

- Water's polarity leads to hydrogen bonding and other unusual properties.
- Atoms in a covalently bonded molecule may share electrons equally, creating a nonpolar molecule. If electrons are shared unequally, a polar molecule is created.
- In a water molecule, oxygen exerts a stronger pull on the shared electrons than hydrogen. This makes the oxygen end of the molecule slightly negatively charged. The hydrogen end of the molecule is slightly positively charged. Both ends act as two different charged poles, known as **Dipole**.
- Water molecules therefore have a weak attraction for each other, with opposite charges , poles, coming together and causing them to behave as if they were "sticky".
- These attractions are not as strong as normal ionic or covalent bonds and are called **hydrogen bonds**.

Figure 2.2 H-Bonding in water molecules



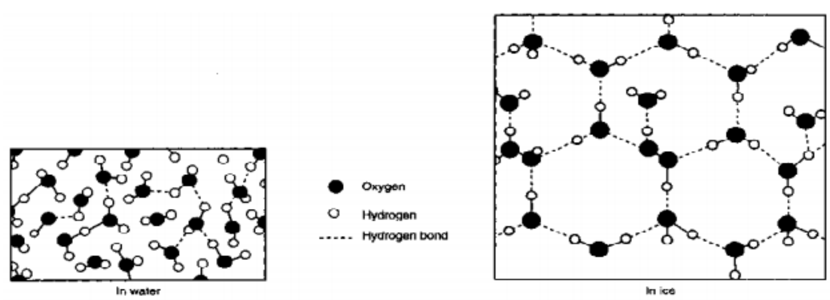


Figure 2.3 H-bonds in Water and Ice

- **Hydrogen bond** is the electrostatic attraction between a positively charged region of one molecule and a negatively charged region of another.
- They are constantly being formed, broken and reformed in water.
- Although individually weak, their collective effect is responsible for many unusual physical properties of water.
- *Why the density of ice is much lower than the water that allows ice cube floated?* The answer lies in the fact that under room temperature, water molecules exist in the form of liquid and only 20% of such hydrogen bonds are formed. But as temperature decreased to 0ºC, all water molecules are involved in forming the three-dimensional structure of ice.

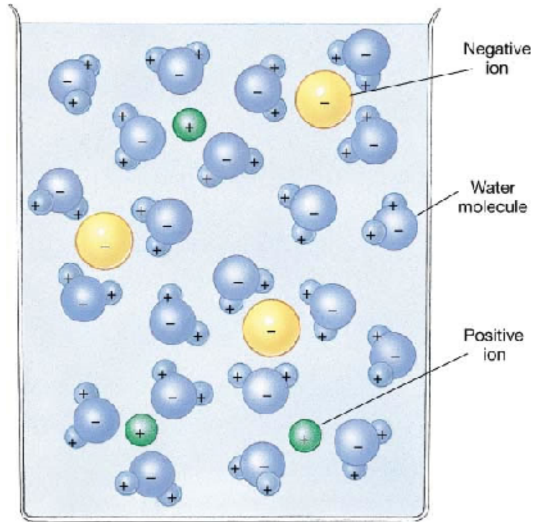
Water as a solvent

- Water is unusual among small molecules by being a liquid at room temperature.
- Water is an <u>excellent solvent</u> for polar substances. These include ionic substances like salts,

which contain charged particles (ions), and some non-ionic substances like sugars hat contain polar groups (slightly charged) such as the slightly negative hydroxyl group (-OH).

- On contact with water, the ions and the polar groups are surrounded by water molecules which separate (dissociate) the ions or molecules from each other. This is happens when a substance dissolves in water.
- Once a substance is in solution its molecules or ions can move about freely, thus making it more chemically reactive than if it were solid. Increased water content leads to increased chemical activity within the cell because all the chemical reactions that take place in cells do so in aqueous solution. Many chemical reactions require water as part of their reactants.
- However, water is a poor solvent for proteins, lipids and other large organic molecules in living cells.

Figure 2.4 Solvent Properties of Water

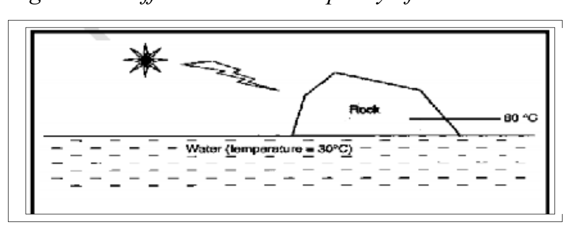


- Non-polar molecules, such as lipids, are repelled by water and usually group together in its presence, that is non –polar molecules are hydrophobic (water-hating).
- Such hydrophobic interactions are important in the formation of membrane and help to determine the 3D structure of many protein molecules, nucleic acids and other cell structures.
- Due to this poor solubility, liquid water can transport nutrient elements to and within living cells without dissolving and destroying the organic molecules of which cells are made.
- Water also transports waste elements away from cells through the extracellular fluid that bathes all cells.
- Water's solvent properties also mean that it acts as a transport medium, as in blood, lymphatic and excretory systems, and the alimentary canal and in xylem and phloem.

High Specific Heat Capacity

- Water's hydrogen bonds can moderate the temperature.
- Water has a high heat capacity. This means that a large increase in heat energy results in a relatively small rise in temperature.
- The specific heat capacity of water is the amount of heat required to raise the temperature of 1kg of water by 1 degree Celcius.
- It takes a lot of energy to disrupt hydrogen bonds. Therefore water is able to absorb a great deal of heat energy without a large increase in temperature. As water cools, a slight drop in temperature releases a large amount of heat.
- Temperature changes within water

Figure 2.5 Difference in heat capacity of water and rock



are minimized as a result of it high specific heat capacity.

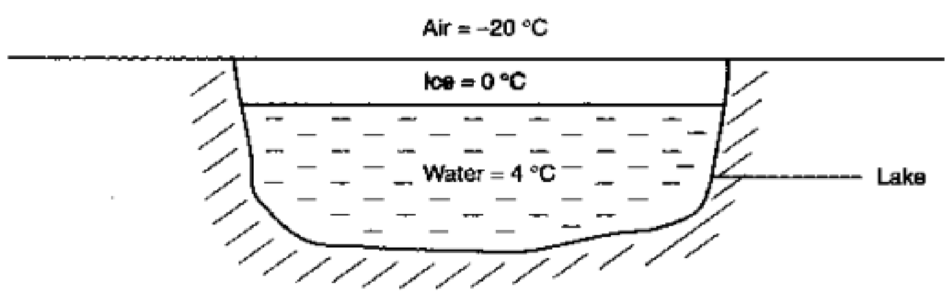
- Biochemical processes therefore operate over smaller temperature range, proceeding at more constant rates and are less likely to be inhibited by extremes of temperature.
- This temperature-stabilizing effect of water is very significant in the cases, where the range of temperatures in which biochemical processes can proceed is quite narrow, and most cells cannot tolerate wide variations in temperature. The presence of water helps keep the temperature of cells fairly stable.

High Heat of Vaporization

- Cooling effect due to heat absorption by water to break H-bond upon vaporization.
- Latent heat of vaporisation is a measure of the heat energy required to vaporise a liquid that is to overcome the attractive forces between its molecules so that they can escape as a gas.
- A relatively large amount of energy is needed to vaporise water. This is due to the hydrogen bonding. As a result, water has an unusually high boiling point for such a small molecule.
- Its significant physiological roles include;
  (a) Enable many invertebrate to survive.
  (b) Cooling effect when we sweat.
  (c) Transpiration helps lower temperature in the plants during hot day.

Density and Freezing Properties

- It is the only substance whose solid form is less dense than its liquid form.
- The density of water decreases below 4ºC and ice therefore tends to float. Since ice floats, it forms at the surface first and the bottom last.
- If ponds froze from the bottom upwards, fresh water life could not exist in temperate or arctic climates. Ice insulates the water below it, thus increasing the chances of survival of organisms in the water.



Unusual Freezing properties of water

- Water is an unusual liquid because when water freezes it increases in volume. Generally, most liquids increase in density, and so decrease in volume, on solidifying.
- Cells that freeze increase in volume so much they normally burst. So does the environment have to be higher than freezing point for a cell to survive?
- It seems not, as nearly all the Antarctic continent is permanently covered with ice but quite a lot of 'lower' plant groups – like moss, fungi, liverworts – and lichens live there. Many marine invertebrates live in the below-freezing temperatures of the Antarctic marine environment.
- ***How is it that these organisms can survive the extreme cold of Antarctica?*** To help answer

this, we can look at how cars are designed to prevent damage to engines in freezing conditions. An automobile's engine is cooled by jackets of water. In winter, water in these channels can freeze and expand. The expansion is so forceful it can crack an engine block. Antifreeze is a liquid solute (ethylene glycol) that dissolves easily in water. In solution and added to a car's engine, it lowers the freezing point of water, preventing it from freezing and ruining the engine.

- **Glucose**, like the antifreeze ethylene glycol, dissolves in water and depresses the freezing point of water. In cells, lowering the freezing point of the liquid protects it from freezing and ripping open with fatal results.
- Antarctic marine invertebrates deal with their situation by accumulating salts and organic compounds, such as glucose and amino acids, which lower the freezing point of the body fluids.

Water's High Surface Tension and Cohesion

Figure 2.7 A water strider supported by surface tension.

- Cohesion is the force whereby individual molecules stick together. At the surface of the liquid, a force called surface tension exists between the molecules as a result of cohesive forces between the molecules. These cause the surface of the liquid to occupy the least possible surface area (ideally a sphere) . Surface tension is the resulting force that causes the surface of water to act as if it has an elastic skin.
- At ordinary temperatures, water has the highest surface tension of any known liquid except mercury, and this is of considerable biological significance.
- Due to the strong cohesive forces, water is incompressible. Because of the incompressible nature of water, plant cells become turgid when filled with water, causing them to swell, providing support for the plant.
- The strong cohesive forces between water molecules play an important part in the movement of water up the xylem in the stems of plants. Were these forces much weaker, trees could not be so tall.
- At a less fundamental level, many small organisms rely on surface tension to settle on water or to skate over its surface. Example: water skaters, night-flying insects. Insects can walk on water due to surface tension created by cohesive water molecules.

Other Properties of Water

- Water forms hydration layers around large charged molecules (e.g. proteins) forming "biological colloids" e.g. cerebrospinal fluid and blood.
- **Reactivity** – Water is a reactant in reactions. E.g. hydrolysis (splitting of water) and dehydration synthesis (bond formation by the removal of water) reactions
- **Cushioning** – Water protects body against physical trauma. E.g. cerebrospinal fluid cushions the brain

3. CARBOHYDATES

- Carbohydrates are biological molecules consist of carbon, hydrogen and oxygen atoms and the basic building block is a sugar molecule. The most common and abundant forms are sugars, fibers, and starches.
- All carbohydrates have the empirical formula $C_nH_{2n}O_n$ or this can be re-written as $C_n(H_2O)_n$ and which shows that they are hydrates of carbon in which the ratio of carbon to hydrogen to oxygen is 1:2:1.
- On the structural basis carbohydrates are **polyhydroxy aldehydes** or **ketones** of substances that yield such compounds on hydrolysis.
- These are most important biological molecule that are energy rich in nature and serve as principal energy resources and structural component (Cellulose & Lignin) of cellular organisms. Being a major energy source, Carbohydrates provide the body with the fuel it needs for physical activity and for proper organ function.
- The best sources of carbohydrates—fruits, vegetables, beans, and whole grains—deliver essential vitamins and minerals, fiber, and a host of important phytonutrients. Also, they are found in a wide array of foods—bread, beans, milk, popcorn, potatoes, cookies, corn and soft drinks.
- Generally, carbohydrates are white, solid sparingly soluble in organic liquids but except for certain polysaccharides, soluble in water unlike lipids. This makes carbohydrates easy to transport around the body (from and to the store). Also, carbohydrates are a lot easier and more rapidly digested so their energy is useful if the body requires energy fast.

3.1 Classification

- Carbohydrates are classified according to the number of simple sugar monomer units present in the molecule.
  - Monosaccharide's
  - Oligosaccharides
  - Polysaccharides

Monosaccharide's:

- The word monosaccharide is derived from mono, meaning "one", and saccharide, meaning "sugar". These are simple sugars having one sugar molecule.
- They cannot be further broken down (hydrolysis) into smaller units. These are colorless, crystalline compounds with sweet taste.
- Based upon the number of carbon atoms in a molecules, they can be classified as:
  1. Trioses: contain 3-carbon atoms Eg. Glyceraldehydes, Dihydroxy acetone
  2. Tetroses: contain 4 carbon atoms Eg. Erythrulose, Erythrose, Threose.
  3. Pentoses: contain 5 carbon atoms Eg., Ribose, Deoxyribose, ribulose, Erythrulose
  4. Hexose: contain 6 carbon atoms Eg., Glucose, fructose, galactose etc.
  5. Heptose: contain 7 carbon atoms Eg., Sedoheptulose

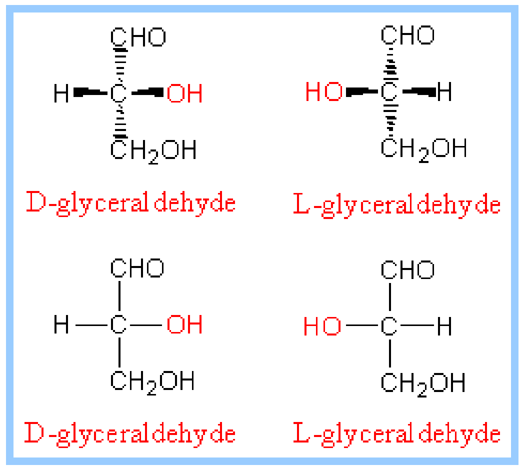
Importance of Monosaccharides

- Pentoses and Hexoses are the most abundant monosaccharides present in protoplasm.
- Glyceraldehyde is the most common form of triose found in the living cells.
- Glucose is the chief source of energy. All carbohydrates are ultimately converted to glucose in the body. Glucose serves as the respiratory substrate in all living cells.
- Ribose is the important component of RNA, NAD, NADP and Acetyl CoA.
- DNA contains Deoxy ribose sugar , which is a pentoses.
- Ribulose is essential for photosynthesis.
- Erythrose is a tetrose involved in synthesis of lignin and anthocyanin.
- Fructose: It is the sweetest of all sugars. It is found in cane sugar, fruit juices, honey and the nectar of flowers.
- Galactose is found commonly in the cells of red algae.

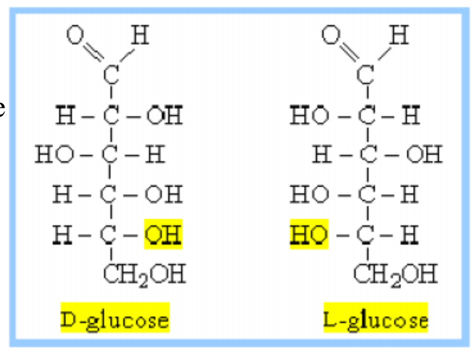
Structure Basics

- Carbohydrates have the basic composition: $(CH_2O)_n$ or H - C - OH

Nomenclature for stereoisomers

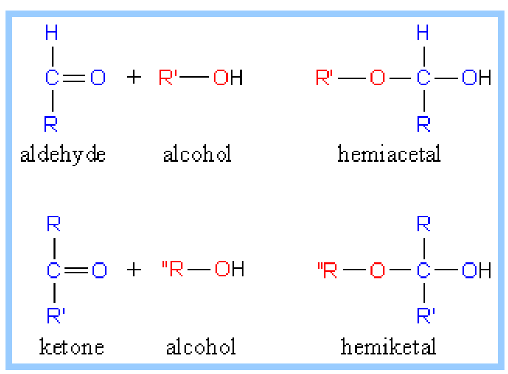


- **D and L designations** are based on the configuration about the single asymmetric carbon in glyceraldehyde.
- For sugars with more than one chiral center, the D or L designation refers to the asymmetric carbon farthest from the aldehyde or keto group.
- Most naturally occurring sugars are D isomers.
- D & L sugars are mirror images of one another. They have the same name. For example, D-glucose and L-glucose.
- Other stereoisomers have unique names, e.g., glucose, mannose, galactose, etc. The number of stereoisomers is 2^n, where n is the number of asymmetric centers. The six-carbon aldoses have 4 asymmetric centers, and thus 16 stereoisomers (8 D-sugars and 8 L-sugars).



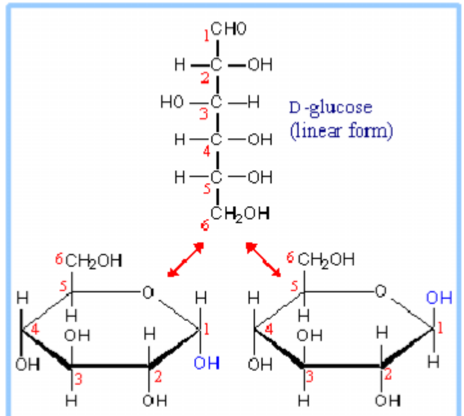
Hemiacetal and hemiketal

- An aldehyde can react with an alcohol to form a hemiacetal.
- Similarly a ketone can react with an alcohol to form a hemiketal.



Fischer and Haworth Projections

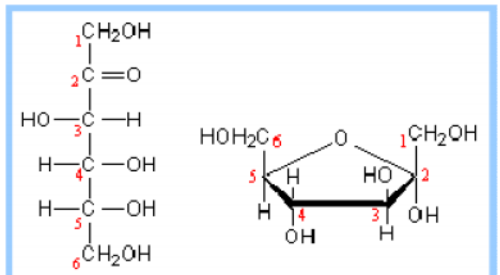
- The representations of the cyclic sugars at right are called **Haworth projections**.
- Pentoses and hexoses can **cyclize**, as the aldehyde or keto group reacts with a hydroxyl on one of the distal carbons.
- E.g., glucose forms an intra-molecular hemiacetal by reaction of the aldehyde on C1 with the hydroxyl on C5, forming a six-member **pyranose** ring, named after the compound pyran.
- Example of **Fischer and Haworth projections**.



α-D-glucose β-D-glucose

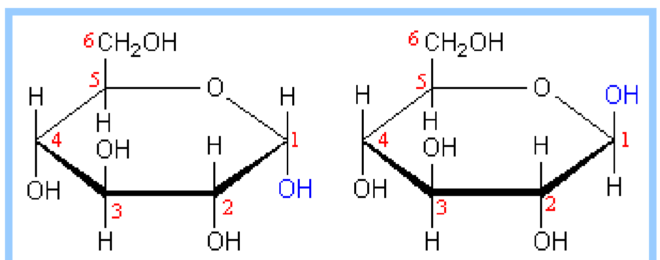
Pyranose ring

- Fructose can form either:
  - a six-member **pyranose** ring, by reaction of the C2 keto group with the hydroxyl on C6
  - a 5-member **furanose** ring, by reaction of the C2 keto group with the hydroxyl on C5.


D-fructose (linear) α-D-fructofuranose

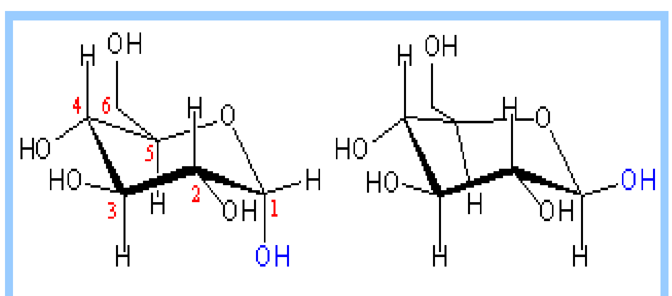
Anomers

- Cyclization of glucose produces a new asymmetric center at C1, with the two stereoisomers called **anomers**, alpha (α) & beta (β).
- Haworth projections represent the cyclic sugars as having essentially planar rings, with the OH at the anomeric C1 extending either:
  - **below** the ring (**a**)
  - **above** the ring (**b**)


α-D-glucose β-D-glucose

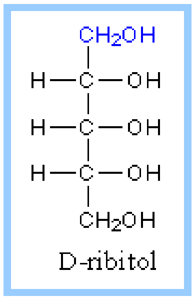
Chair or Boat configuration

- Because of the tetrahedral nature of carbon bonds, the cyclic form of pyranose sugars actually assume a **"chair"** or **"boat"** configuration, depending on the sugar.
- The representation at right reflects the chair configuration of the glucopyranose ring more accurately than the Haworth projection.

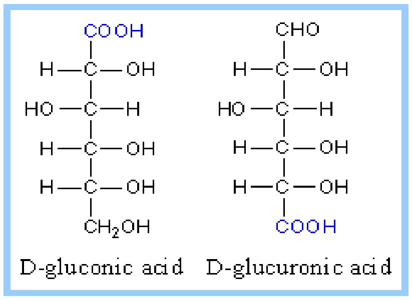

α-D-glucopyranose β-D-glucopyranose

Sugar derivatives

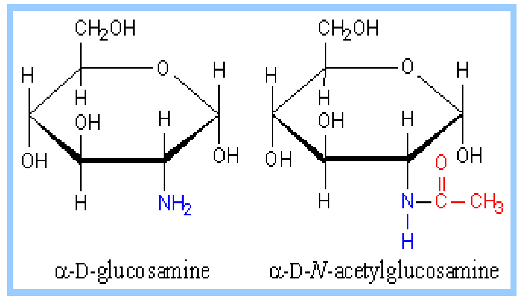
- Various derivatives of sugars exist, including:
- **Sugar alcohol** - lacks an aldehyde or ketone. An example is ribitol.
- **Sugar acid** - the aldehyde at C1, or the hydroxyl on the terminal carbon, is oxidized to a carboxylic acid. Examples are gluconic acid and glucuronic acid.
- **Amino sugar** - an amino group substitutes for one of the hydroxyls. An example is glucosamine. The amino group may be acetylated. At right, the acetic acid moiety is shown in red.



D-ribitol

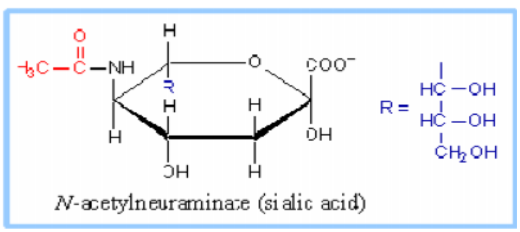


D-gluconic acid D-glucuronic acid



α-D-glucosamine α-D-*N*-acetylglucosamine

- N-acetylneuraminate, (N-acetylneuraminic acid, also called sialic acid) is often found as a terminal residue of oligosaccharide chains of glycoproteins. Sialic acid imparts negative charge to glycoproteins, because its carboxyl group tends to dissociate a proton at physiological pH, as shown here.



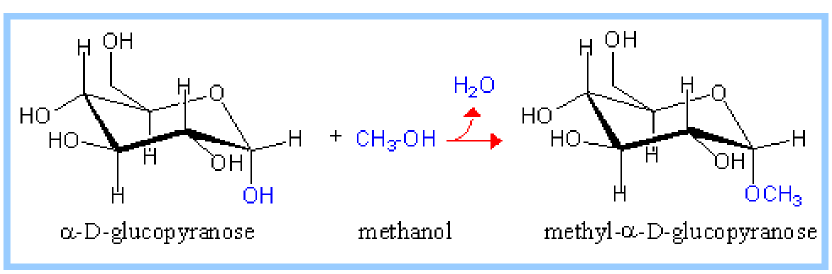
N-acetylneuraminate (sialic acid)

Glycosidic bonds:

- The anomeric hydroxyl group and a hydroxyl group of another sugar or some other compound can join together, splitting out water to form a glycosidic bond.

  R-OH + HO-R' --> R-O-R' + H_2O

- For example, methanol reacts with the anomeric hydroxyl on glucose to form methyl glucoside (methyl-glucopyranose)
-



α-D-glucopyranose methanol methyl-α-D-glucopyranose

Oligosaccharides

- They are carbohydrates formed by the joining of 2 to 6 monosaccharide molecules.
- The bond formed between monosaccharides, is called glycosidic bond. The condensation process takes place between carbonyl group (>C=O) of one monosaccharide unit and hydroxyl group(-OH) of another residue with the elimination of water molecule results to form glycosidic linkage. For example, If the carbon-1 (carbonyl group) of one α-monosaccharide molecule and carbon-4 of the other α-monosaccharide molecule are bonded to get α-1,4 (or α(1→4)) glycosidic bond.
- A large number of oligosaccharides have been prepared by partially breaking down more complex carbohydrates (polysaccharides).
- They can be hydrolyzed by acid to form constituent monosaccharide units.
- Oligosaccharides are classified according to the number of monosaccharide units (or monomers) they are composed of, into disaccharides, trisaccharides, tetrasaccharides, pentasaccharides and hexasaccharides. Among these, disaccharides are most abundant in the living system.

Disaccharides

- The three most important disaccharides are sucrose, lactose and maltose.
- The monomers are very specific. Meaning you must have stereochemistry exact; the bond will require an alpha or beta and always the D form.
- Disaccharides are formed through a dehydration reaction.
- This reaction releases a water molecule.
- To break this bond, named a glycosidic bond, you add water, this reaction is named hydrolysis.

Importance of disaccharides

- Maltose: It is made of two glucose molecules. It is formed from starch during germination of barley or wheat. It is commonly called malt sugar.
- Lactose is a disaccharide found in milk and hence commonly called milk sugar. It is formed by condensation of a glucose and a galactose.
- Sucrose: It is a disaccharide found extensively in plants. It is commonly called cane sugar. It is formed by a condensation of a molecule of glucose and a molecule of fructose.
- Raffinose is a trisaccharide formed by condensation of a molecule each of glucose, fructose and galactose. It occurs in plants.

Figure 2.3 Structure of Disaccharides

Sucrose → Glucose + Fructose (non-reducing sugar)

$$C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6 + C_6H_{12}O_6$$

Lactose → Galactose + Glucose (reducing sugar)

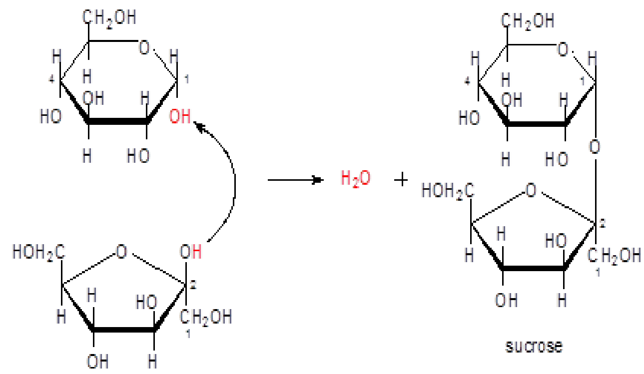
$$C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6 + C_6H_{12}O_6$$

Maltose → Glucose + Glucose (reducing sugar)

$$C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6 + C_6H_{12}O_6$$

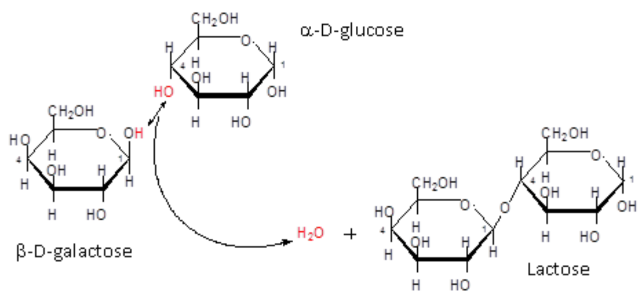
Sucrose:

- most common disaccharide, table sugar
- 20% of sugar cane is sucrose
- it is composed of one α-D-glucose and one β-D-fructose monomer
- linked α-1 → β-2, this linkage information tells you that the α-D-glucose molecule uses its #1 carbon, whose OH group is down, to bond to the #2 carbon on the β-D-fructose molecule, whose OH group is up



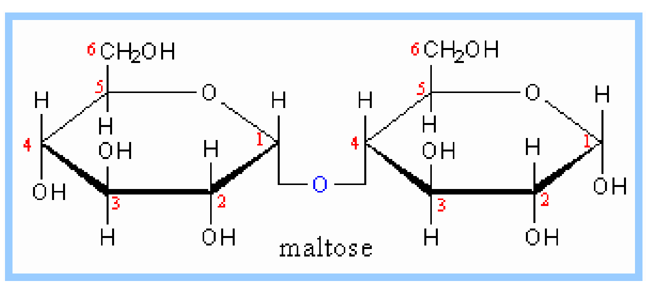
Lactose:

- milk sugar
- by mass composes 7% of human milk, 5% of bovine milk
- being lactose intolerant is a fairly common condition, especially in adults who drink less milk, the body forgets how to make the enzyme, lactase, needed to break lactose's glycosidic bond



- it is composed of one α-D-glucose or β-D-glucose and one β-D-galactose monomer
- linked β-1→ 4
- this linkage information tells you that the α-D-glucose molecule uses its #4 carbon to bond to the #1 carbon on the β-D-galactose molecule, both of which have a OH group in the up position
- the dehydration of one α-D-glucose and one β-D-galactose monomer to form water and lactose

Maltose:



- malt sugar
- it is composed of α-D-glucose monomers
- in an acidic solution maltose undergoes hydrolysis, the resulting solution containing glucose
- linked α-1→ 4 glycosidic linkage.
- this linkage information tells you that one α-D-glucose molecule uses its #1 carbon to bond to the #4 carbon on the other α-D-glucose molecule, both of which have a OH group in the down position

Polysaccharide

- **Polysaccharides** are polymeric carbohydrate structures, formed of repeating units (either mono- or di-saccharides) joined together by glycosidic bonds.
- These structures are often linear, but may contain various degrees of branching.
- Depending on the structure, these macromolecules can have distinct properties from their monosaccharide building blocks. They may be amorphous or even insoluble in water.
- Polysaccharides contain more than ten monosaccharide units. Definitions of how large a carbohydrate must be to fall into the categories polysaccharides or oligosaccharides vary according to personal opinion
- When all the monosaccharides in a polysaccharide are the same type the polysaccharide is called a ***homopolysaccharide***, but when more than one type of monosaccharide is present they are called ***heteropolysaccharides***.
- Polysaccharides is an important class of biological polymers. Their function in living organisms is usually either **structure**- or **storage**-related.

Storage polysaccharides

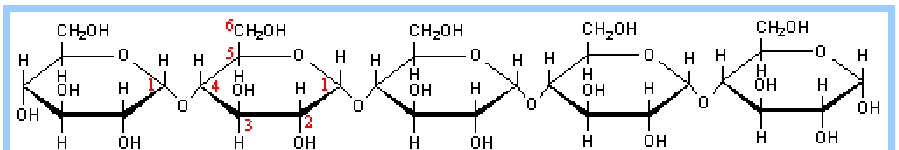
- Starch (Plants)
- Glycogen (Animals)

Structural polysaccharides

- Cellulose
- Chitin
- Cellulose and chitin are examples of a glycan. Glycans may be homopolymer or heteropolymer depending on their monosaccharide residue components. Glycans may also be linear or branched.

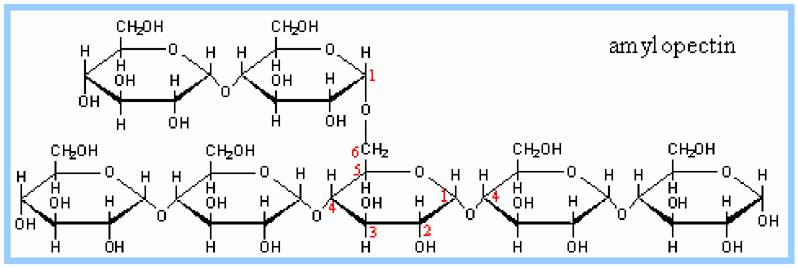
Starch

- Starches are glucose polymers in which glucopyranose units are bonded by *alpha*-linkages. It is made up of a mixture of **Amylose** (15–20%) and **Amylopectin** (80–85%).
- Starches are insoluble in water. It is stored in **chloroplasts** of **plants**.
- Starch is a digestible polysaccharide. They can be digested by hydrolysis, catalyzed by enzymes called **amylases**, which can break the *alpha*-linkages (glycosidic bonds).
- Humans and other animals have amylases, so they can digest starches. Potato, rice, wheat, and maize are major sources of starch in the human diet. The formation of starches are the way that plants store glucose.
- **Amylose** consists of a linear chain of several hundred glucose molecules with alpha **(1,4)** glycosidic linkages. The end of the polysaccharide with an anomeric carbon (C1) that is not involved in a glycosidic bond is called the **reducing end**.



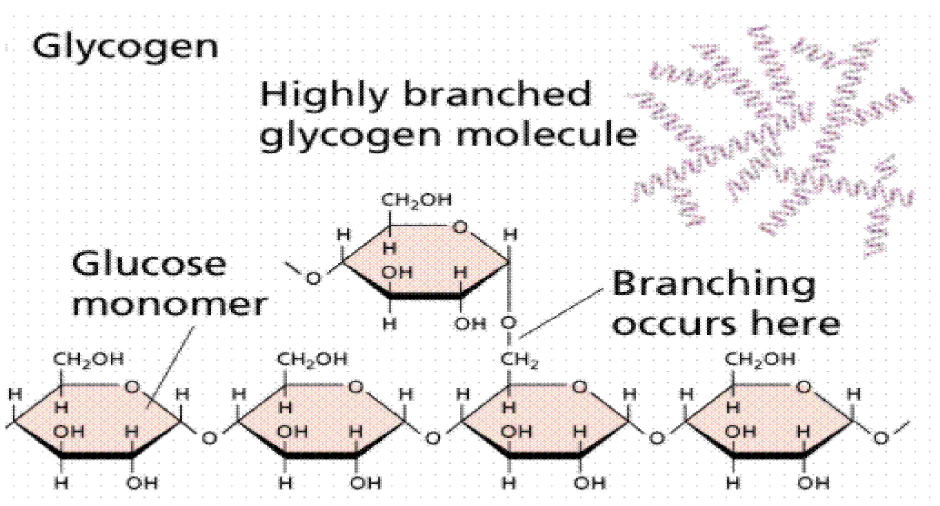
amylose

- **Amylopectin** is a glucose polymer with mainly alpha **(1, 4)** linkages, but it also has **branches** formed by alpha **(1, 6)** linkages. The branches are generally longer than shown above. The branches produce a compact structure, and provide multiple chain ends at which enzymatic cleavage of the polymer can occur. It is made of several thousand glucose units (every chain 24–30 glucose unit).



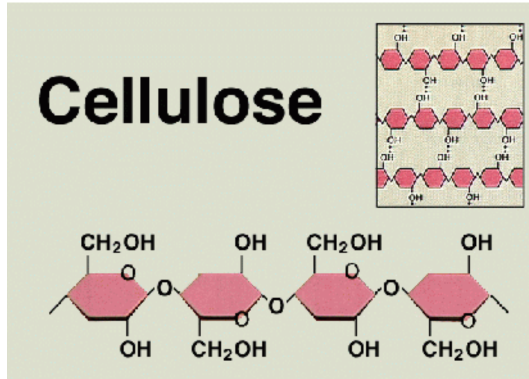
Glycogen

- Glycogen is composed of a branched chain of glucose residues and serves as energy reserve for animals. Also chief form of carbohydrate stored in animal body.
- It is insoluble in water. It also yields glucose on hydrolysis.
- Glycogen serves as the secondary long-term energy storage in animal and fungal cells, with the primary energy stores being held in adipose tissue.
- Glycogen is the analogue of starch, a glucose polymer in plants, and is sometimes referred to as animal starch, having a similar structure to amylopectin (with alpha acetal linkages) but more extensively branched and compact than starch.
- Glycogen is a glucose polymer of α(1→4) glycosidic bonds linked, with α(1→6)-linked branches. Glycogen has **more alpha (1,6) branches** as compared to starch.
- Various samples of glycogen have been measured at 1,700-600,000 units of glucose.
- The highly branched structure permits rapid release of glucose from glycogen stores, e.g., in muscle cells during exercise. The ability to rapidly mobilize glucose is more essential to animals than to plants. Glycogen is made primarily by the liver and the muscles, but can also be made by glycogenesis within the brain and stomach.

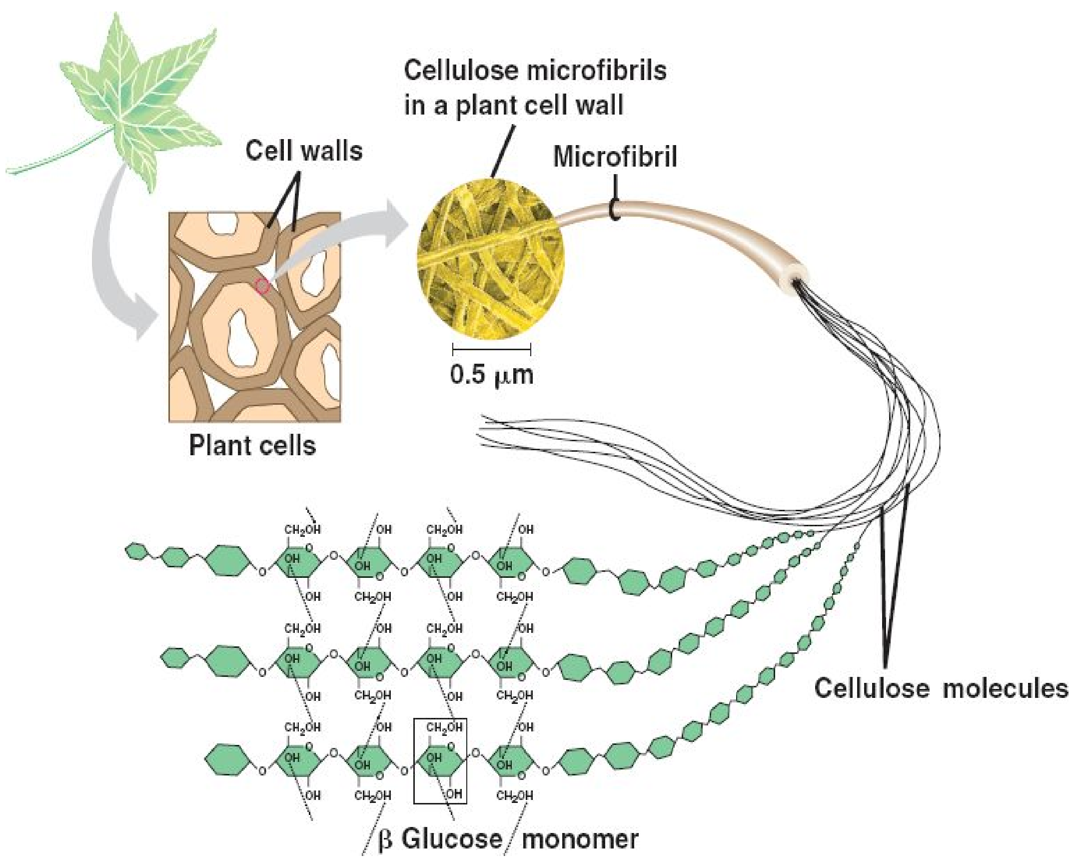


Cellulose

- Cellulose, a major constituent of plant cell walls, consists of long linear chains of glucose, with alpha (1, 4) linkages.
- Parallel cellulose molecules are held together by hydrogen bonds between hydroxyl groups **of glucose monomers.**
- Every other glucose in cellulose is flipped over, due to the glycosidic linkages. This promotes intrachain and interchain hydrogen bonds, as well as van der Waals interactions, that cause cellulose chains to be straight and rigid, and pack with a crystalline arrangement in thick bundles called **microfibrils**.

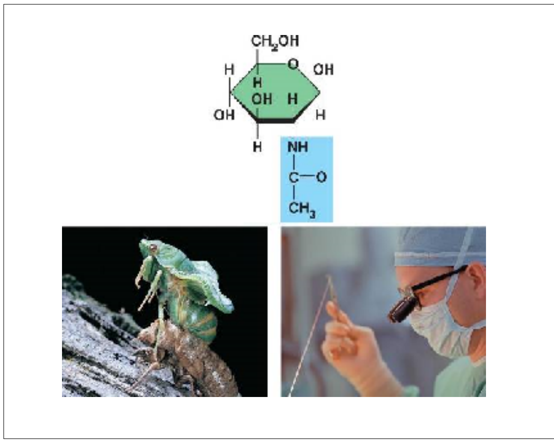


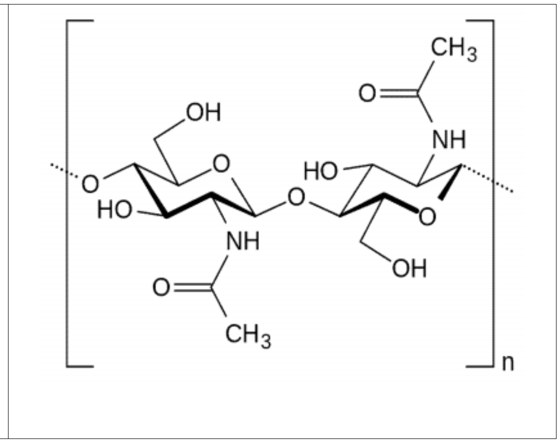
- Cellulose, also known as **plant fiber**, cannot be digested by human beings and so cellulose passes through the digestive tract without being absorbed into the body.
- Some animals, such as cows and termites, contain bacteria in their digestive tract that help them to digest cellulose.
- Cellulose is a relatively stiff material, and in plants cellulose is used as a structural molecule to add support to the leaves, stem and other plant parts.



Chitin

- Chitin is one of many naturally occurring polymers. It forms a structural component of many animals, such as **exoskeletons**.
- **Chitin** is made of glucose monomers whose 2 carbon has a **nitrogen** group.
- Chitin is also used to make biodegradable **surgical threads**.
- Chitin can be broken down by an enzymes called **chitinases**, which is secreted by microorganisms such as bacteria and fungi, and also produced by some plants.
- The chitase enzyme cleaves the <u>glycosidic bonds</u> and results into simple sugars and ammonia.
- It is also closely related to cellulose in that it is a long unbranched chain of glucose derivatives.





Pectin

- Pectins, also known as pectic polysaccharides, are rich in galacturonic acid. Several distinct polysaccharides have been identified and characterised within the pectic group. Homogalacturonans are linear chains of α-(1–4)-linked D-galacturonic acid.
- Pectin is an important cell wall polysaccharide that allows primary cell wall extension and plant growth. Pectin is a major component of the middle lamella, where it helps to bind cells together, but is also found in primary cell walls.
- Pectin is a natural part of the human diet, but does not contribute significantly to nutrition. The daily intake of pectin from fruits and vegetables can be estimated to be around 5g (assuming consumption of approximately 500 g fruits and vegetables per day).
- In human digestion, pectin binds to cholesterol in the gastrointestinal tract and slows glucose absorption by trapping carbohydrates. Pectin is thus a soluble dietary fiber.
- The main use for pectin (vegetable agglutinate) is as a gelling agent, thickening agent and stabilizer in food.
- In medicine, pectin increases viscosity and volume of stool so that it is used against constipation and diarrhea.
- Pectins are mainly polygalacturonic acids with differing degrees of D-galactosyl, L-arabinosyl, or L-rhamnosyl residues.
- Pectins are predominant in the middle lamella, the layer between neighbouring cells.

4. LIPIDS

- Lipids are the largest and most diverse group of naturally occurring organic compounds composed of carbon, hydrogen and oxygen.
- They are soluble in non-polar organic solvents like ether, chloroform, acetone & benzene, and generally insoluble in water.
- Lipids are a group of molecules which include: fats, oils, waxes, phospholipids, steroids (like cholesterol), sphingolipids, prostaglandins and fat-soluble vitamins.
- Lipids form about 3.5% of the total chemical composition of a cell.
- Lipids can be either amphipathic (having polar and nonpolar qualities) or hydrophobic (nonpolar).
- Unlike carbohydrates and proteins, lipids are highly polymorphic and structurally difficult to define and classify.
- Lipids do not form large bio-polymers like other biomolecules can form because they do not share a common molecule structure or repeating structural unit. Like Carbohydrates (monosaccharides units), Proteins(amino acids units) and Nucleic Acids(nucleotides units).
- However some lipids are long chained molecules because of long aliphatic hydrocarbons tail as a part of fatty acids.

Biological importance of lipids

- Lipids act as a source of energy and are used as metabolic fuels. Lipids in adipose tissue serve as energy store. Adipose tissue fat-stores contain 9.45 kcal/g (= ~37 kJ/g) of dry weight lipid. This compares to ~4 kcal/g for protein or carbohydrate. Thus, they yield twice the energy produced by the same weight of carbohydrates or proteins.
- They are the natural solvent for fat-soluble vitamins, thus provide means whereby fat-soluble nutrients (e.g., sterols, vitamins) can be absorbed by the body. Lipids include the vitamins A, D, E, and K, and include some non-vitamin enzyme cofactors.
- Lipids serve as structural element of cell and sub-cellular components. They are primarily involve in the formation of cell membranes which is a bilipid layer and present in all living things.
- Some lipids (such as sterols) also act as hormones like estrogens, testosterone, and progesterone, cortisol which are chemical messengers enter in cells and turn on specific chemical reactions.
- Lipids act as **insulation**, and play a role in physical appearance of animals. Lipids under the skin serve as thermal insulator.
- Lipids in myelin sheath of nerve fibers serve as electrical insulator.
- **Amphipathicity** :Most lipids are amphipathic: they have both polar and non-polar characteristics. The polar regions of lipids are capable of forming favorable interactions with water, while the non-polar regions tend to aggregate with other non-polar molecules. This differential polarity is critical for their structural role in membranes, as well as in a number of functional roles.
- Some lipids such as Lipoproteins are essential components in the structure of cell membrane and mitochondria. Also, they are important for lipid transport in the blood.

Classification of Lipids

Lipids are classified into three groups:

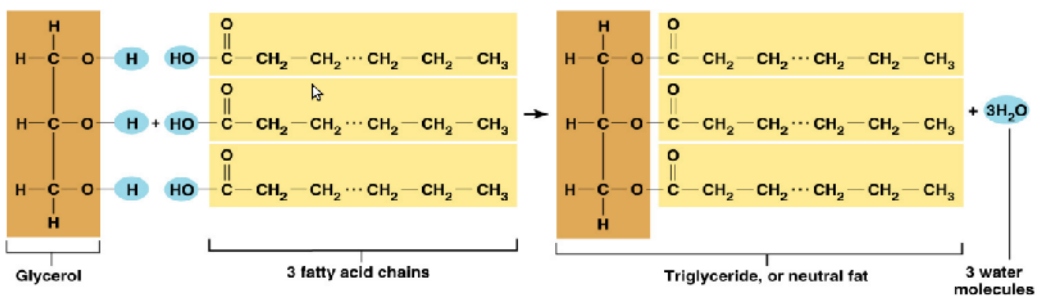
1. **Simple lipids**
   - They are esters of fatty acids. For example: fats and oil, wax.
   - Fats are esters of fatty acids and glycerol.
   - Waxes are esters of long chain fatty acids and long chain alcohols.
   - On hydrolysis yield at most two types of primary product per mole

2. **Complex (conjugated) lipids**
   - These are lipids, which contain an inorganic or organic group in addition to fatty acids and alcohol (or glycerol).
   - Thus, composed of simple lipids plus other non-lipid moities.
   - Includes Phospholipids, Glycolipids, Sulpholipids and Lipoproteins
   - They yield three or more primary hydrolysis products per mole.

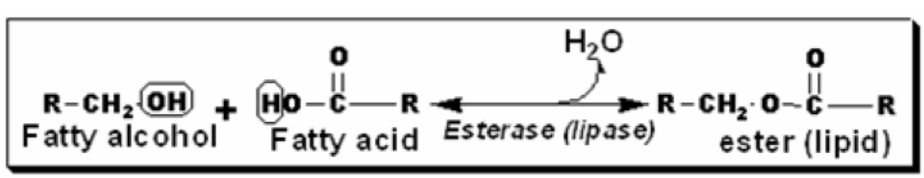
3. **Derived lipids**
   - include those that cannot be neatly classified into either of the above, means include other water insoluble substances such as steroids (cholesterol, bile acids and hormones), eicosanoids (prostagladins) and isoprenoids including fat-solubles vitamins (A,D,E,K).
   - Also includes compounds derived from simple and complex lipids (by hydrolysis).
   - Composed of hydrocarbon rings and a long hydrocarbon side chain.

Simple lipids

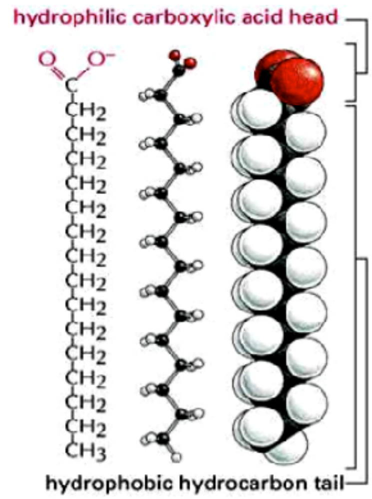
- In general, simple lipids are the ester of fatty-acid with alcohol.
- The simplest lipids are the fatty acids with formula R-COOH, where R is hydrocarbon tail.
- Simple lipids group includes fat, oil and wax.
- The most common simple lipid in nature consist of fatty acids linked by an ester bond to the trihydric alcohol (glycerol) to form Triacylglycerols (or Triglycerides).



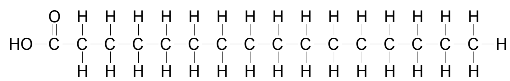
(a) Formation of a triglyceride



Fatty Acids



- Carboxylic acids having terminal polar hydrophilic "head" (carbonyl group) and long non-polar hydrophobic "tail" of hydrocarbons.
- Fatty acids are aliphatic mono-carboxylic acids that are mostly obtained from the hydrolysis of natural fats and oils.
- In lipids, it is usually attached to glycerol or other backbone such as cholesterol etc.
- Have general formula R-(CH2)n-COOH and mostly have straight chain (a few exceptions have branched and heterocyclic chains). In this formula "n" is mostly an even number of carbon atoms (2-34) with a few exceptions that have an odd number.
- The most abundant saturated fatty acids are palmitic (C16) and stearic (C18)



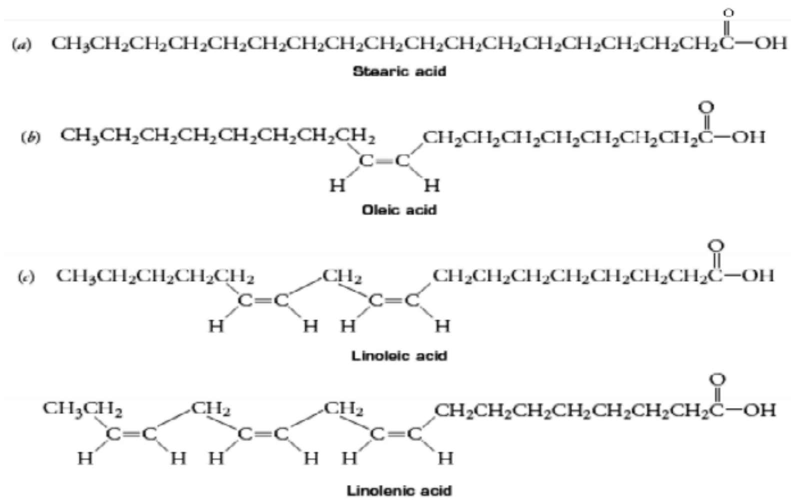
palmitic acid - a fatty acid

Table: Common Fatty Acids

Common name	# of C atoms	# of = bond	Tm (°C)	Formula	Source
Acetic acid	2	0		CH3-COOH	Butter fat
Butyric acid	4	0		CH3-(CH2)2-COOH	Butter fat
Caproic acid	6	0		CH3-(CH2)4-COOH	Butter fat
Caprylic acid	8	0		CH3-(CH2)6-COOH	Coconut oil
Capric acid	10	0		CH3-(CH2)8-COOH	Coconut oil
Lauric acid	12	0	43	CH3-(CH2)10-COOH	Coconut oil
Myristic acid	14	0	54	CH3-(CH2)12-COOH	Nutmeg
Palmitic acid	16	0	62	CH3-(CH2)14-COOH	Palm
Parmitoleic acid	16	1	-0.5	CH3-(CH2)5 -CH=CH-(CH2)7-COOH	Macadamia
Stearic acid	18	0	69	CH3-(CH2)16-COOH	Lards
Recinoleic acid	18	1	14	CH3-(CH2)7 -CH=CH-(CH2)7-COOH	Caster Oil
Oleic acid	18	1	13	CH3-(CH2)7 -CH=CH-(CH2)7-COOH	Olives
Linoleic acid	18	2	-9	CH3(CH2)4CH=CH(CH2)CH=CH(CH2)7COOH	Safflower
Linolenic	18	3	-17	CH3(CH2)CH=CH(CH2)CH=CH(CH2)CH=CH(CH2)7COOH	Flaxseed oil
Arachidic acid	20	0	77	CH3-(CH2)18-COOH	Peanut,fish
Arachidonic acid	20	4	-50	CH3(CH2)4-(CH=CH-CH2)4(CH2)2-COOH	Peanut,fish
EPA (Eicosapantaenoic acid)	20	5	-50	CH3(CH2)-(CH=CH-CH2)5(CH2)2-COOH	Fish oil
Erucic acid	22	1		CH3-(CH2)7 -CH=CH-(CH2)11-COOH	Rapeseed oil
DHA (docosahexanoic)	22	6		CH3(CH2)-(CH=CH-CH2)6-(CH2)-COOH	Fish oil

Naming conventions of fatty Acids

- Specify chain length by number
- Then, colon (:) followed by number of double bonds
- Then , n , position of double bonds
    - Palmitic acid: $CH_3(CH_2)_{14}COOH$ = C16:0
    - Stearic acid: $CH_3(CH_2)_{16}COOH$ = C18:0
    - Oleic acid: $CH_3(CH_2)_7CH=CH(CH_2)_7COOH$ = C18:1,n-9
    - Linoleic: $CH_3(CH_2)_4$ **$(CH=CH-CH_2)_2$**$(CH_2)_6COOH$ = C18:2, n-9, 12.
    - Linolenic acid: $CH_3(CH_2)$ **$(CH=CH-CH_2)_3$**$(CH_2)_6COOH$ = C18:3, n-9, 12, 15.
    - Arachidonic acid $CH_3(CH_2)_4$ **$(CH=CH-CH_2)_4$**$(CH_2)_2COOH$ = C20:4, n-5,8,11,14.

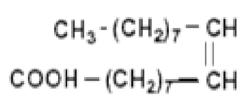
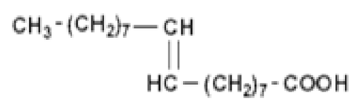


Classification of Fatty acids

- Fatty acids vary in carbon chain lengths, degree of unsaturation, and location of double bonds. Saturation affects the physical characteristics of the fat and its storage properties.

    - **On the basis of carbon chain length**
        - **Short chain fatty acids**
            - less than 10 carbons
            - liquid at room temperature
            - water-soluble
            - volatile
            - Found in dairy products
            - Examples: Acetic acid, butyric acid and caproic acid
        - **Long chain fatty acids**
            - More than 10 carbons
            - solid at room temperature
            - Water insoluble
            - Nonvolatile
            - Found primarily in meat, fish, and vegetable oils.
            - Examples: Palmetic and Stearic acid

- **On the basis of degree of unsaturation**
  - **Saturated fatty acids**
    - They have no double bonds between carbon atoms
    - carry the maximum possible number of hydrogen atoms
    - They have the general formula **$CH_3 – (CH_2)_n$ - COOH**
  - **Unsaturated fatty acids**
    - They have one or more double bonds between carbon atoms
    - The double bond is considered the **point of unsaturation**.
    - On the basis of no. of double bonds they are further classified as
      - **Monounsaturated fatty acids**
        - have only one double bond.
        - Examples: Oleic acid (C18:1,n-9)
      - **Polyunsaturated fatty acids (PUFA)**
        - have at least two or more double bonds.
        - Examples: Linoleic acid**,** Linolenic acid, Arachidonic acid etc.
    - ***Cis* and *Trans* configuration:** They may be present in cis and trans stereoisomeric forms
      - *Cis* configuration means that the groups around the double bond are on the same side of the bond.
      - *Trans* configuration means that the groups around the double bond are on the opposite sides of the bond.
      - For example: Oleic acid (C18:1,n-9), which is a cis form, and its isomer is called eliadic acid , which exists as trans form.

	
Cis Oleic acid	*Trans* eliadic acid

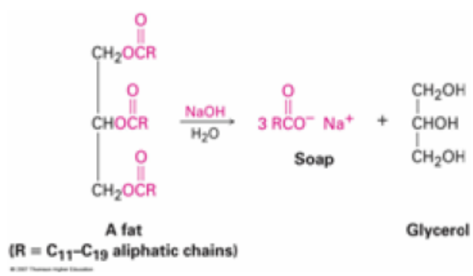
- **On the basis of location of double bonds**
  - The **omega** number refers to the position of the first double bond.
  - An **omega-3 fatty acid** has the location of the double bond in the third position. An example is linolenic acid.
  - An **omega-6 fatty acid** has the location of the double bond in the sixth position. An example is linoleic acid.

Essential fatty acids

- human body is capable of synthesizing most fatty acids from carbohydrates or other fatty acids. However, humans do not synthesize sufficient amounts of fatty acids that have more than one double bond, called as **Essential fatty acids**.
- These fatty acids are called essential fatty acids because they must be provided by the diet. They are essential for growth.
- They are polyunsaturated fatty acids i.e. fatty acids which contain more than one double bond between carbon atoms.
- They include linoleic, linolenic and arachidonic acids.

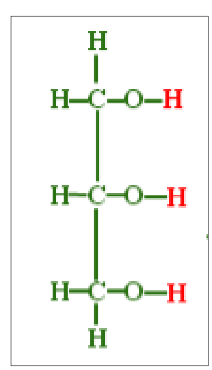
Physical and Chemical Properties of fatty acids

- **Physical Properties**
  - They are colourless, odourless and tasteless.
  - **Solubility**: They are insoluble in water with the exception of short-chain fatty acids which are soluble in water. The solubility decreases with the increase in chain length. All fatty acids are soluble in organic solvents
  - **Melting point:** It depends upon the length of the chain and degree of unsaturation.
    - Short-chain and unsaturated fatty acids have low melting points. They are liquid at room temperature
    - Long chain saturated fatty acids have higher melting point. They are solid at room temperature.
  - **Stereoisomerism**: Unsaturated fatty acids can be present in cis and trans stereoisomeric forms. Cis-fatty acids are found naturally while trans can be synthesized artificially by partial hydrogenation of fats and oils. For example Oleic (*Cis* form) and Eliadic acids (*Trans* form)
- **Chemical Properties**
  - **Saponification or Salt formation** (Soaps): Being acids, fatty acids react with alkali to form salts, called **Soaps**. The method of alkaline hydrolysis is known as **saponification**. Sodium and potassium soaps are soluble in water so they are called soft soaps. Calcium and magnesium soaps are insoluble in water so they called as hard soaps.



  - **Ester Formation** (Reaction with alcohols) : Fatty acids react with alcohol to form esters. For example, fatty acids react with glycerol to form mono, di and triglycerides.
  - **Hydrogenation**: Addition of hydrogen to unsaturated fatty acids at the double bond changes it to the corresponding saturated fatty acids
  - **Reduction**: Fatty acids can be reduced to fatty aldehydes, fatty alcohols, and hydrocarbons.
  - **Oxidation** : Unsaturated fatty acids are oxidized by oxygen when exposed to air forming lipid peroxides, fatty aldehydes, ketones and short chain fatty acids

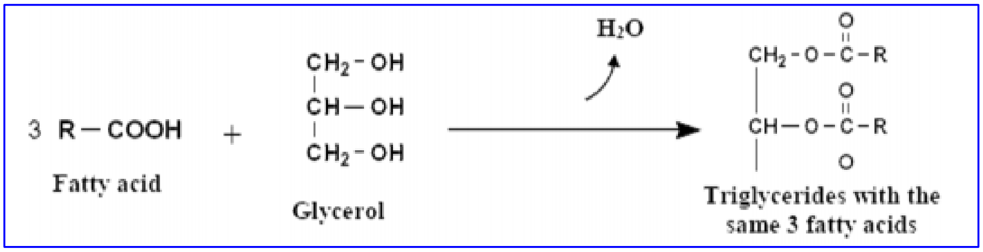
Glycerol



- It is a trihydric alcohol.
- It is colourless and viscid fluid with sweet taste.
- It is miscible with water in all proportions.
- With strong dehydrating agents, as concentrated sulphuric acid, glycerol can be converted to acrolein that has very irritating odour. This is called **acrolein test**.
- Importance of glycerol:-
  - It is used in pharmaceutical and cosmetic preparations.
  - It is used as explosive in the form of trinitroglycerine.
  - It is used in medicine as a vasodilator agent in coronary heart diseases in the form of nitroglycerine.

Fats and Oils (Triacylglycerol)

- They are esters of fatty acids with glycerol. They are similar chemically, but they differ in physical properties, as oils are liquid while fats are solid at room temperature.
- Fat , oil and wax are made from two kinds of molecules:
    - one **glycerol** (a type of alcohol with a hydroxyl group on each of its **three** carbons)
    - and three **fatty acids** joined by esterification, a dehydration reaction.
- Triacylglycerols are the major form of energy storage for animals.



- A triacylglycerol is produced by esterification (a reaction in which the hydroxyl groups of glycerol form ester bonds with the carboxyl groups of fatty acids.
- Most fats and oils are mixed triacylglycerols that contain two or three different fatty acids.

Waxes

- A wax is an ester of a saturated fatty acid and a long-chain alcohol, each containing from 14 to 30 carbon atoms. Mostly they are esters of acids with long chain monohydric alcohol.
- Waxes are found in many plants and animals. Coating of carnauba wax on fruits and the leaves and stems of plants help to prevent loss of water and damage from pests.
- Waxes on the skin, fur, and feathers of animals and birds provide a waterproof coating.
- They are solid at room temperature.
- Example: Triacontyl hexadecanoate is in beeswax, cholesterol esters in blood.

Triacontyl hexadecanoate (from beeswax)

Differences between fats and waxes

	Fats and oils	**Waxes**
Consistency at room temperature	Fats are solid. Oils are liquid	Solid
Structure	Glycerol and 3 fatty acids	Long chain alcohol and fatty acid
Glycerol	Present	Absent
Long chain alcohol	Absent	Present
Acrolein test	Positive	Negative
Rancidity	They can undergo rancidity	Do not undergo rancidity
Digestion	Digested by lipase	Not digested
Utilization by human body	Can be utilized	Can not be utilized

Rancidity

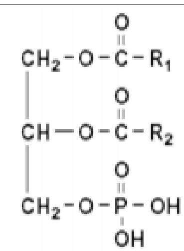
- Rancidity is a condition in which fat attains a bad taste and disagreeable odour.
- Rancidity is predisposed by Light, Moisture and Warm temperature.
- Rancidity leads to:
  - Fats and oils attain bad taste.
  - Fats and oils attain disagreeable odour.
  - Production of toxic compounds as lipid peroxides, aldehydes and ketones.
- Rancidity can be prevented by:
  - Addition of antioxidants to fats and oils specialty the natural antioxidant tochopherol (vitamin E).
  - Also, avoid exposure of fats to light, moisture and high temperature.

Compound (Conjugated) Lipids

- Compound (conjugated) lipids are lipids conjugated with other substances which could be an inorganic or organic group in addition to fatty acids and alcohol (or glycerol).
- They include:
  1. Phospholipids formed of lipid and phosphate moiety.
  2. Glycolipids, formed of lipid part and carbohydrate part
  3. Sulpholipids, lipids containing sulphate.
  4. Lipoproteins formed of lipid part and protein part

Phospholipids

- Lipids which contain a phosphate moiety are grouped together as phospholipids.
- This group of lipids is derived from **phosphatidic acid**, a molecule with glycerol esterified to two fatty acids and one molecule of phosphoric acid.
- The fatty acids most commonly found in phosphatidic acid are palmitic and stearic acids in their saturated forms.
- The further esterification of phosphatidic acid with a low-molecular-weight alcohol (i.e. ethanolamine, choline, serine, or inositol) gives rise to other complex phospholipids.



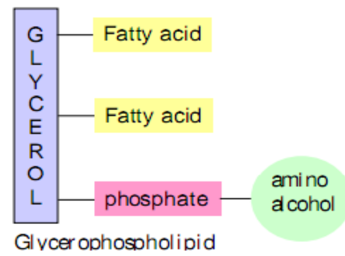
Phosphatidic acid

Types of Phospholipids

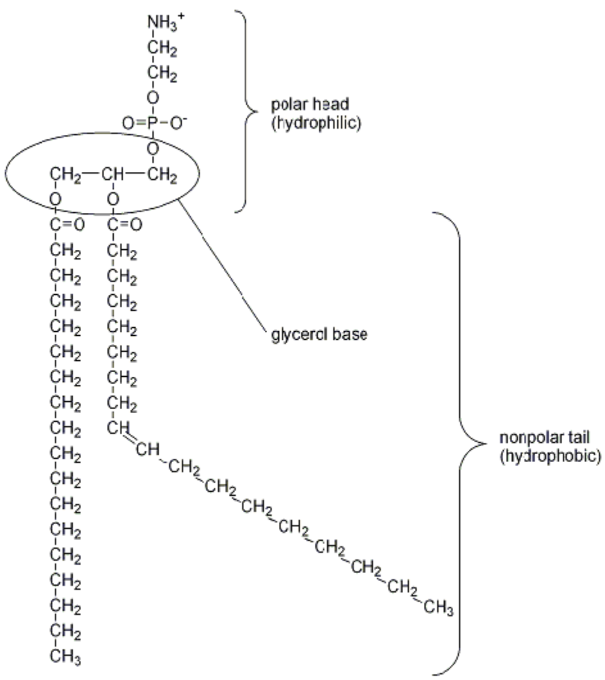
On the basis of the alcohol present with Phophatidic acid, they are classified into two groups

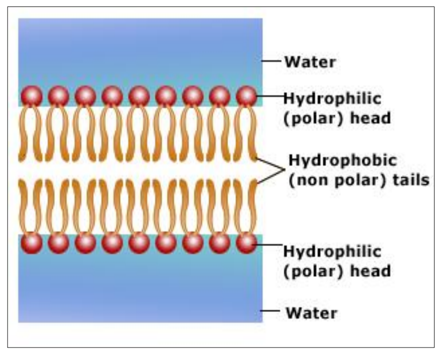
- Glycero-phospholipids
  - Alcohol is glycerol or other low-molecular-weight alcohols like ethanolamine, choline, serine and inositol.
- Sphingomyelin
  - Alcohol is sphingol alcohol or sphingosine

Glycero-phospholipids

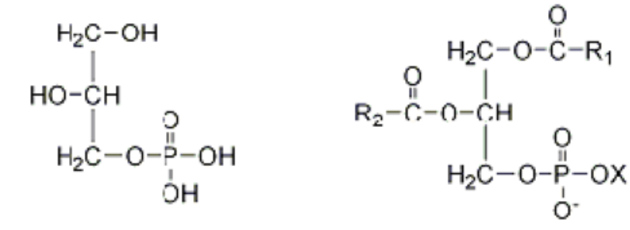


- Glycerophospholipids (also called phosphoglycerides) are the structural components of **lipid bilayers.**
- Phosphoglycerides are similar to triacylglycerols except that one hydroxyl group of glycerol is replaced by the ester of phosphoric acid and amino alcohol, bonded through a phosphodiester bond.
- Three **amino alcohols** found in glycerophospholipids are **choline**, **serine**, and **ethanolamine**.
- The main reason phospholipids are of such great interest is their presence in membranes.
- The glycerophospholipds are amphipathic molecules with polar heads and (usually) long nonpolar tails.
-

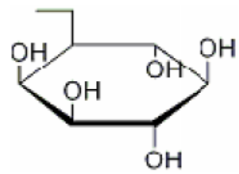




- This lipid bilayer is the basis of all cell membrane structures, and acts as a relatively impermeable barrier to the passage of most water soluble molecules.
- Protein molecules embedded in this bilayer can act as transporters for molecules that would normally not be able to pass through the membrane.
- Bilayers form spontaneously among phospholipids molecules in aqueous environments due to their amphipathic nature. When surrounded by water on all sides, they aggregate to protect their hydrophobic tails in the interior of the bilayer or micelle.
- The bilayer is actually the more favorable structure of the two since it protects the tails on all sides from interaction with water.
- The most commonly found structures of phospholipids in nature are as follows:



sn-glycerol-3-phosphate　　　phospholipids

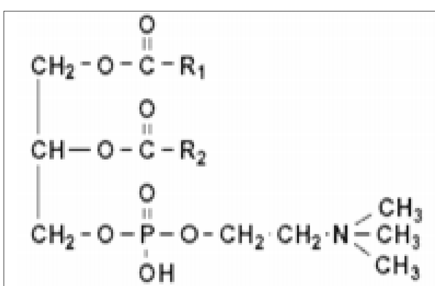
name of X-OH	formula of X	name of phospholipid
water	$-H$	phosphatidic acid
ethanolamine	$-CH_2CH_2NH_3^+$	phosphatidylethanolamine
choline	$-CH_2CH_2N(CH_3)_3^+$	phosphatidylcholine
myoinositol		phosphatidylinositol
glycerol	$-CH_2CH(OH)CH_2OH$	phosphatidylglycerol

Common name of phospholipids with low-molecular-weight alcohols are:

- Lecithin - phosphatidyl cholines
- Cephalin - phosphatidyl ethanolamines and phosphatidyl serines
- lipoinositol - Phosphatidyl inositol

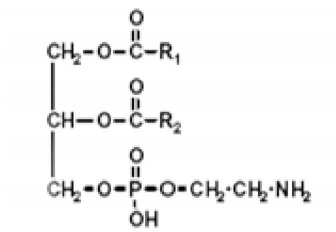
Structure of Lecithin

- It is formed of:
  - Glycerol base
  - Saturated fatty acid, attached to alpha carbon of glycerol by ester bond.
  - Unsaturated fatty acid, attached to beta carbon of glycerol by ester bond.
  - Phosphoric acid attached to alpha carbon of glycerol by ester bond
  - Choline attached to phosphoric acid by ester bond
- Food sources of **lecithin** include eggs, liver, soybeans, wheat germ and peanuts.

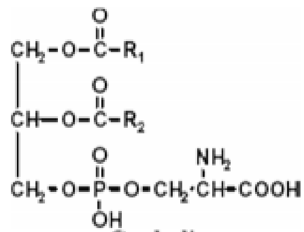


Lecithin (phosphatidyl choline)

Cephalins: They are phosphatidyl ethanolamine and phosphatidyl serine



Cephalin
(Phosphatidyl ethanolamine)

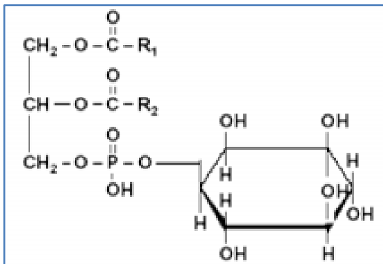


Cephalin
(Phosphatidyl serine)

- They have a role in blood coagulation.
- They accelerate blood clotting because they enter in the structure of thromboplastine, which is essential for blood clotting.

Phosphatidyl inositol

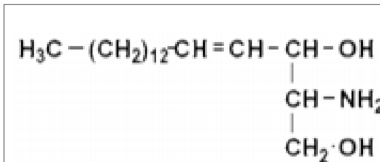
- It is also called lipoinositol. It is found in brain tissue.
- It has a role in mechanism of hormone action.



Phosphatidyl inositol

Sphingomyelin

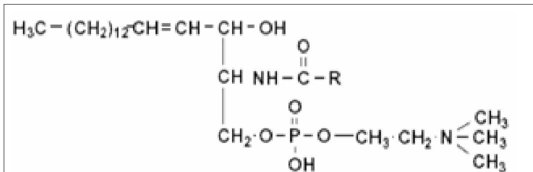
- In this type of phospholipids the alcohol is sphingol alcohol which is also called sphingosine base i.e. sphingomyelin contains sphingosine base instead of glycerol.
- Sphingosine base contains 18 carbon atoms:
  - The first carbon contains hydroxyl group (- OH).
  - The second carbon contains amino group (-NH2)
  - The third carbon contains hydroxyl group.
  - There is a double bond between C4 and C5



Sphingosine base

Structure of sphingomyelin

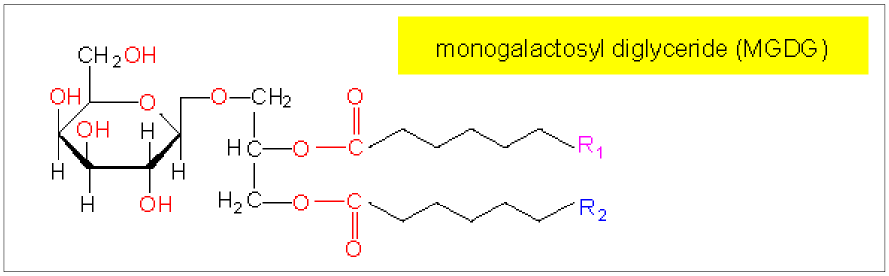
- Sphingomyelin is formed of:
  - Sphingosine base
  - Unsaturated fatty acid attached to the amino group of sphingosine
  - Phosphoric acid attached to the first carbon of sphingosine.
  - Choline base attached to phosphoric acid.
- It is abundant in the nervous system in the myelin sheath also it is present to lesser extent in liver, spleen and bone marrow.

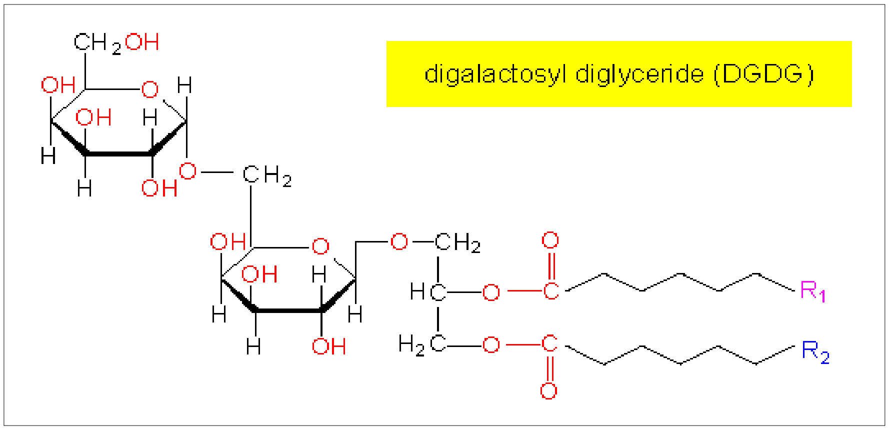


Sphingomyelin

Glycolipids

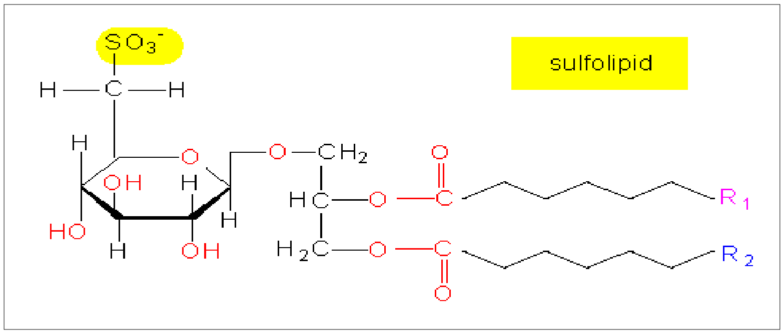
- formed of lipid part and carbohydrate part
- In **glycolipids,** the glycerol-residue carries a sugar instead of a phosphate. They are characteristic for plastid membranes.





Sulpholipids

Sulpholipids, are similar in structural with glycolipids except a sulphate ion is attached to the sugar molecule.



Lipoproteins

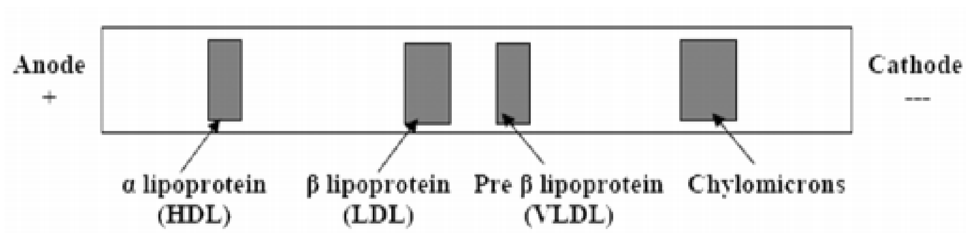
- These are compound lipids formed of lipid part (which may be triglycerides, cholesterol or phospholipids) and protein part (which may be alpha or beta globulin).

Function of lipoproteins

- They enter in the structure of cell membrane. They are also found in milk and egg yolk.
- They are important for lipid transport in the blood. Lipids are insoluble in water so they cannot be transported alone. Lipids bind to protein to make lipoproteins are water-soluble and can be transported in the blood.

Classification of lipoproteins

- Lipoproteins are classified into 4 types according to the rate of floatation in sodium chloride solution after ultracentrifugation. These types are chylomicrons, very low-density lipoproteins (VLDL), low-density lipoproteins (LDL) and high-density lipoproteins (HDL).
- Also, lipoproteins are classified into 4 types according to migration in electric field (electrophoresis). These types are chylomicrons, pre-beta lipoproteins, beta lipoproteins and alpha lipoproteins.



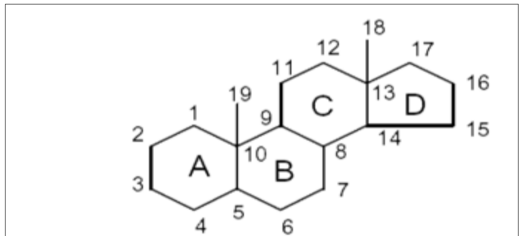
	Chylomicrons	**VLDL**	**LDL**	**HDL**
Site of synthesis	Small intestine	Liver	Blood	Liver
Structure	Mainly triglycerides, small amount of cholesterol, phospholipids and protein	Principally triglycerides, greater amount of cholesterol, phospholipidss and protein	Mainly cholesterol and appreciable amount of proteins, phospholipids and triglycerides	Mainly of proteins and phospholipids, small amount of cholesterol and little amount of triglycerides
Size	Largest	Large	Small	Smallest
Rate of floatation	Highest	High	Low	Lowest
Electrophoresis	Do not migrate because they have no charge	Migrate before globulin	Migrate with globulin	Migrate with globulin

Derived Lipids

- They include those substance that cannot be neatly classified into either simple or complex lipids. They includes:
  - Steroids (cholesterol, bile acids and hormones),
  - Eicosanoids (prostagladins) and
  - Isoprenoids including fat-solubles vitamins (A,D,E,K).
  - Also includes compounds derived from simple and complex lipids (by hydrolysis).

Steroids

- Steroid are compounds containing the steroid nucleus, which consists of **three cyclohexane** rings and **one cyclopentane** ring fused together (no fatty acids). The four rings in the steroid nucleus are designated A, B, C, and D.
- The carbon atoms are numbered beginning with the carbons in ring A and, in types of steroids like cholesterol, ending with two methyl groups (attached groups make a wide variety of steroid compounds).
- They are a large group of biologically important compounds. All of them contain a steroid nucleus, which is cyclopentano-perhydrophenanthrene nucleus.

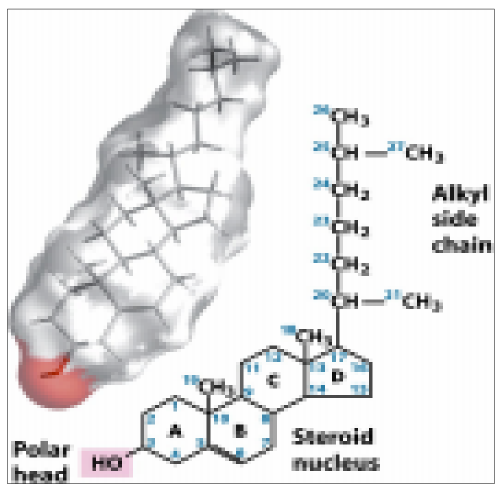


Cyclopentano-perhydrophenanthrene (steroid) nucleus

- Steroid include: Cholesterols, Steroid hormones, Steroid Vitamins (Vitamins D) and Bile acids

Cholesterol

- **Cholesterols**: is one of the most important and abundant steroids in the body. Cholesterol has methyl groups, alkyl chain and hydroxyl group (-OH) attached to the steroid nucleus.
- Cholesterol is a component of cellular membranes, myelin sheath,and brain and nerve tissue.
- It is also found in the liver, bile salts, and skin, where it forms vitamin D.
- Cholesterol in the body in obtained from eating meats, milk, and eggs, and it is also synthesized by the liver from fats, carbohydrates, and proteins. There is no cholesterol in vegetable and plant products. If the diet is high in cholesterol, the liver produces less. We should consume not more than 300 mg of cholesterol a day.



Eicosanoids

- The lipids called eicosanoids are hormones that are derived from arachidonic acid, a polyunsaturated fatty acid containing 20 carbons (the prefix eico means 20).
- When hydrolyzed from a certain phospholipid by hormone action, arachidonic acid undergoes reactions that tranforms it into the various eicosanoids - prostaglandins, thromboxanes, and leukotrienes.

Prostaglandins

- The prostoglandins are derived from the oxidation of arachidonic acid.
- The members of this group of structurally related natural hormones have an extraordinary range of biological effects.
    - They can cause pain, inflammation and fever
    - They can lower gastric secretions, stimulate uterine contractions and induce labor (PGE2), lower blood pressure, influence blood clotting and induce asthma-like allergic responses.

Thromboxanes

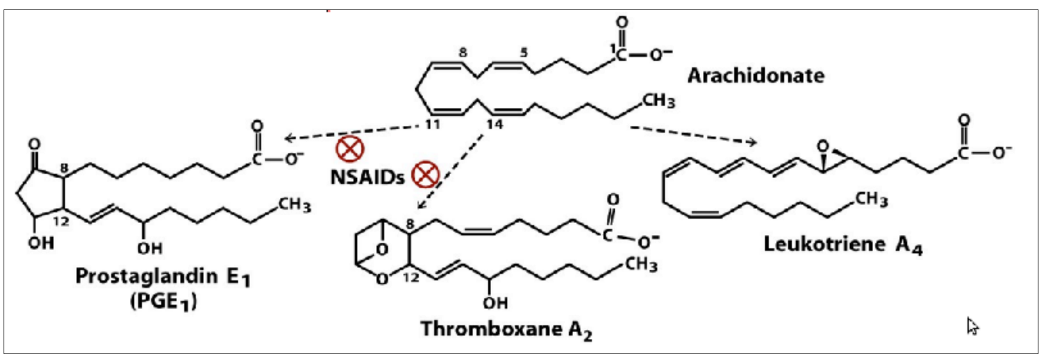
- Thromboxanes, such as thromboxane A2, are involved in platelet aggregation and blood clotting. When a blood vessel is compromised thromboxanes are released causing the platelets to clot the leak. Blood thinning medication, such as aspirin will reduce the ability of platelets to come together and can cause excessive bleeding, especially during surgery.

Leukotrienes

- They are mainly found in white blood cells. They can produce long lasting muscle contractions which can cause asthmatic attacks.Some anti-asthma drugs block the production of leukotrienes.

Nonsteroidal Anti-inflammatory Drugs (NSAIDs)

- NSAIDs such as aspirin, acetaminophen, and ibuprofen reduce pain, fever, and inflammation by blocking the action of an enzyme involved in the conversion of arachidonic acid into prostaglandins and thromboxanes. There are two forms of this enzyme: COX-1 and COX-2



5. PROTEINS

- Proteins are organic molecules consisting of many amino acids bonded together.
- Proteins are made up of polypeptides. Polypeptides are a chain of amino acids connected together.
- The properties of a protein are largely determined by its three-dimensional structure.
- Amino acids have the general formula NH_2-CH(R)-COOH, where R is anything (e.g. if R = H then the amino acid is glycine, if R = CH_3 then the amino acid is alanine.).
- Polypeptides are made up of combinations of 20 different amino acids . Glycine, alanine, valine, leucine, isoleucine, methionine, phenylalanine, tyrosine, tryptophan, aspartic acid, glutamic acid, lysine, arginine, histidine, asparagine, glutamine, serine, threonine, cysteine and proline.
- The nature of the 'R' group determines how hydrophilic or hydrophobic the amino acid is.
- Alanine, valine, leucine, isoleucine and methionine have aliphatic, non-polar side chains and are therefore hydrophobic. Phenylalanine, tyrosine and tryptophan have aromatic (ring) side chains and are very hydrophobic. The rest are hydrophilic.
- A protein may be composed of a single polypeptide or it may be made up of several subunits (e.g. the a, b and g subunits of G-proteins).
- Any chain <25 amino acids is called an oligopeptide, a chain containing >25 amino acids is a polypeptide. The largest polypeptide contains 5000 amino acids. Most polypeptides contain less than 2000.
- The polypeptide will always have NH2 at one end, the **N terminus** and COOH at the other, the **C terminus**.

Important Functions of Proteins with Named Examples

Enzyme catalysis -
- faciliates/speeds up certain chemical reactions; eg. enzymes
- Amylase converts starch to simple sugar.

Defense -
- recognizes foreign microbes; forms the center of the immune system;
- eg. immunoglobulins, toxins, antibodies (Globular proteins that "recognize" foreign microbes.)

Transport -
- moves certain small molecules/ions; eg. hemoglobin, proton pump
- Hemoglobin is red blood cell protein.

Structure/ support -
- structural role; eg. fibers, collagen (most abundant protein in vertebrates), keratin, fibrin
- Collagen, which forms the matrix of skin, ligaments, tendons and bones.

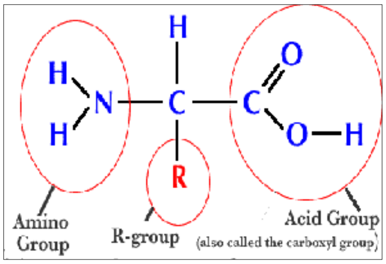
Motion -
- a muscle protein responsible for muscle contraction
- contracting muscles; eg. actin, myosin

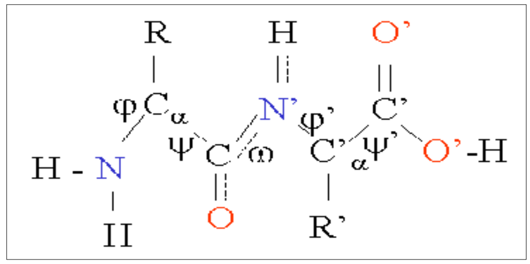
Regulation -
- receives/sends information to regulate body functions; ex. hormones
- Hormones which serve as intercellular messengers; eg. Insulin (blood sugar regulation).

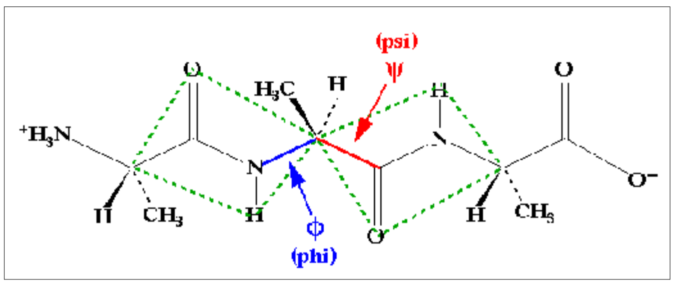
Storage -
- holds molecules such as calcium and iron; ex. ferritin

Amino Acids and Peptide Bond

- Monomers or building blocks of all proteins.
- Parts of the Amino Acid:
  a) <u>Amino group</u> (NH2) and b) <u>Carboxyl group</u> (COOH)
  c) R-group: variable- 20 R-groups, so only 20 amino acids.
- **Dipeptide:** 2 amino acids joined by peptide bond.
- **Polypeptide:** many amino acids bonded together by peptide bond.







Amino acid Properties

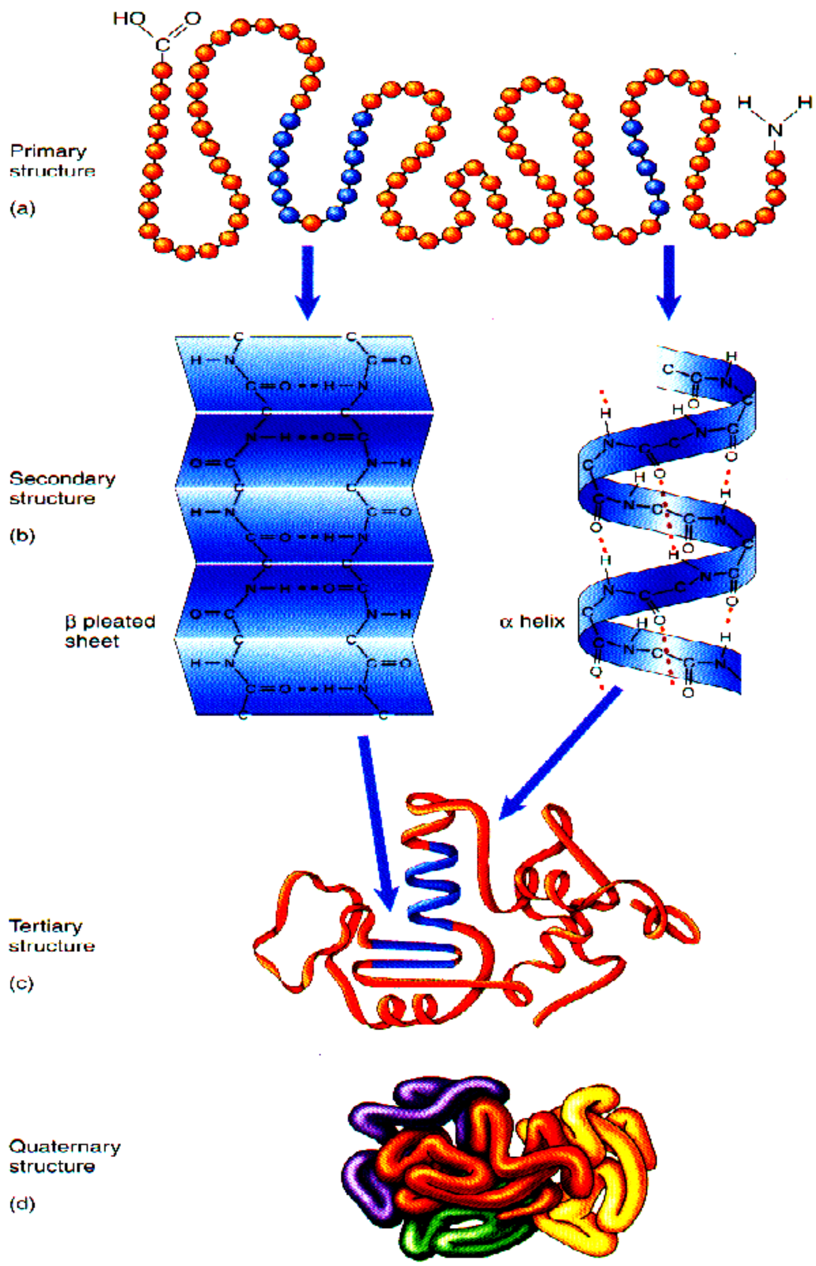
- 20 different kinds used in specific orders to form proteins
- molecule consists of an amino group, carboxyl group, hydrogen atom, and side group (determines the molecule's characteristics) connected to a central carbon atom
- nonpolar amino acids have CH2 or CH3 as side group
- polar amino acids have oxygen or hydrogen as side group
- charged amino acids have acids/bases as side group
- aromatic amino acids have organic rings w/ alternating single/double bonds as side group
- special-function amino acids have unique individual characteristics
- **peptide bond** - bonds between amino acids; forms between the hydrogen and carboxyl groups

Peptide bonds:

- Bond formed when 2 amino acids bond by condensation synthesis
- The peptide group has a rigid, planar structure which is a consequence of resonance interactions that give the peptide bond a ~40% double-bond character.
- Peptide groups, with few exceptions, assume the trans conformation. Successive Ca atoms are on opposite sides of the peptide bond joining them.
- Polypeptide backbone conformations may be described by their torsion angles
- Allowed conformations are indicated by the Ramachandran diagram
- Only three small regions of the conformational map are physically accessible to a polypeptide chain.

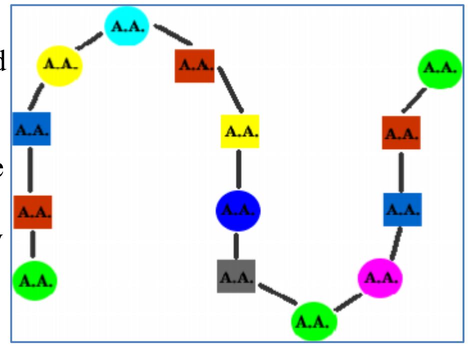
Protein structure - shape determines function

- Shape found through x-ray diffraction or NMR.
- Internal amino acids are generally nonpolar
- Most polar/charged amino acids are found on the surface
- Different levels of structure - primary, secondary, motifs, tertiary, domains, quaternary
- **Factors of protein shape** - hydrogen bonds between amino acids, disulfide bridges between side chains, ionic bonds, Van der Waals attractions (weak attractions due to electron clouds), hydrophobic exclusion (polar portions gather on the outside, nonpolar portions go towards the interior).



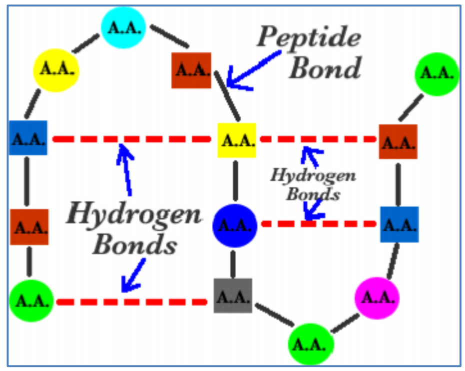
Levels of Protein Structure

1) **Primary structure:**



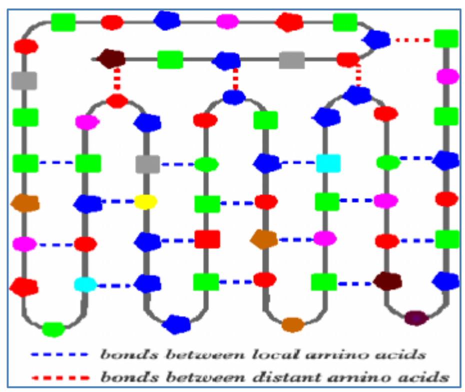
- Single polypeptide chain with amino acids connected by peptide bonds. (No H-bonds between nearby amino acids!)
- determined by nucleotide sequence that codes for the protein
- any of the 20 different amino acids can appear at any position in a protein
- side groups play no role in peptide structure, but important in primary structure

2) **Secondary structure:**



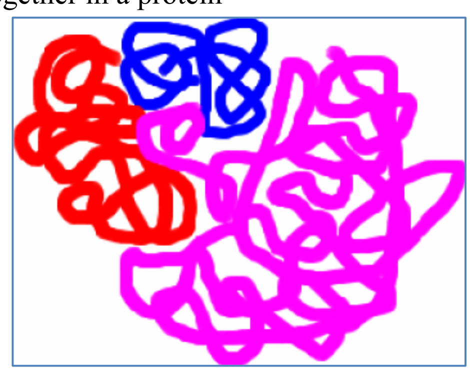
- Single polypeptide chain. H bonding to side groups of amino acids several places away, forms folded or coiled structure.
- A polymer's secondary structure (2' structure) is defined as the local conformation of its backbone.
- **Secondary protein structure** - determined by hydrogen bonds folds the amino acid chain
- Alpha helix - forms when hydrogen bonds form in a chain
- Beta helix - when parallel chains are linked into a pleated shape

3) **Tertiary structure:**



- Single polypeptide chain. 3-D shape due to H-bonds between amino acids that is nearby as well as distant!
- final folded shape of the globular protein
- protein goes into the tertiary form due to hydrophobic exclusion
- can be unfolded (denatured) and still return to original shape
- no holes in the protein interior
- close nonpolar chains are attracted together by van der Waal's forces
- change in any amino acid can affect how they stay together in a protein

4) **Quaternary structure:**



- two or more polypeptide chains linked to form complex 3-D structure.
- quaternary structure - 2+ polypeptide chains connecting to form a functional protein
- arrangement of the subunits
- subunits connect to each other in nonpolar areas
- altering a single amino acid can affect the entire structure

Denaturation - unfolding of proteins

- can occur if pH, temperature, or ionic concentration is changed
- leads to biologically inactive proteins (venoms, made of proteins, stop working in high temperature or in presence of acids/bases)
- salt-curing/pickling used high concentrations of salt/vinegar to stop the enzymes of microorganisms from working
- most enzymes can only function well in very specific conditions
- usually, only smaller proteins can fully refold themselves after being denatured
- **dissociation** - different from denaturation; subunits can dissociate and still go back to their quaternary structure

Protein Classification

A. Classification of Proteins According to Shape and Solubility

Proteins can be broadly classified into three groups, based on their shape and solubility.

a. **Fibrous proteins**: these proteins have a rod like structure. They are not soluble in water. Collagen is an example of a fibrous protein.

b. **Globular proteins**: these proteins more or less spherical in nature. Due to their distribution of amino acids (hydrophobic inside, hydrophillic outside) they are very soluble in aqueous solution. Myoglobin is an example of a globular protein.

c. **Membrane proteins**: these are proteins which are in association with lipid membranes. Those membrane proteins that are embedded in the lipid bilayer have extensive hydrophobic amino acids that interact with the non-polar environment of the bilayer interior. Membrane proteins are not soluble in aqueous solution. Rhodopsin is an example of a membrane protein. Note that rhodopsin is an integral membrane protein and is embedded in the bilayer. The lipid membrane is not shown in the structure presented here.

B. Classification of Globular Proteins According to Secondary Structure

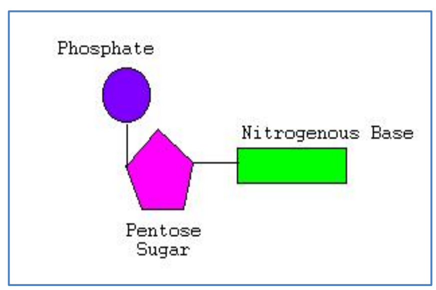
Proteins are also classified based on the type of secondary structure present.

a) **All alpha**: Proteins that contain only (in some exceptional instances there may be isolated beta-sheets) alpha helical secondary structure. Myoglobin is an example of an all alpha protein.
b) **All beta**: Protein that contain only (in some exceptional instances there may be isolated alpha helices) beta-sheet secondary structure. Tenascin is an example of an all beta protein.
c) **Alpha/beta**: Proteins that contain alternating alpha-helical and beta-sheet secondary structure elements. Triose Phosphate Isomerase is an example of an alpha/beta protein. This structure is known as a TIM barrel. The alternating helical and sheet segments form a closed barrel like structure.
d) **Alpha + Beta**: In these proteins the alpha helical and beta sheet regions occur in independent regions of the molecule. Ribonuclease A is an example of an alpha+beta protein.

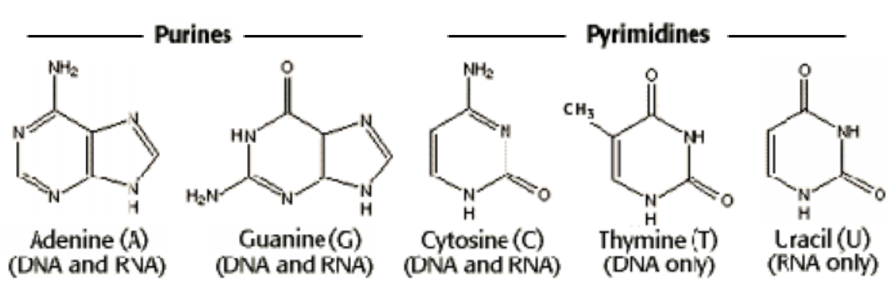
6. Nucleic acids

- Nucleic acids were discovered by Friedrich Miescher in 1869.
- Nucleic acids are the most important biological macromolecules which is essential for all living organisms. They function in encoding, transmitting and expressing genetic information.
- Nucleic acids constitute the genetic material present in all living systems. This genetic material carries the information about the reconstruction of living organism which is in the form of a code. This code is essential to regulate and propagate life.
- ***Types:***
    - They are two types of nucleic acids:
        - **Deoxyribose nucleic acid (DNA):** Double stranded contain deoxy ribose sugar.
        - **Ribose nucleic acid (RNA) :** Single stranded contain ribose sugar.
- ***Size:***
    - Nucleic acids can vary in size, but are generally very large molecules.
    - Nucleic acid molecules range in size from 21 nucleotides (small interfering RNA) to large chromosomes (human chromosome 1 is a single molecule that contains 247 million base pairs).
- ***Biopolymer***:
    - Nucleic acid is the largest biopolymer consists of monomeric units called nucleotides.

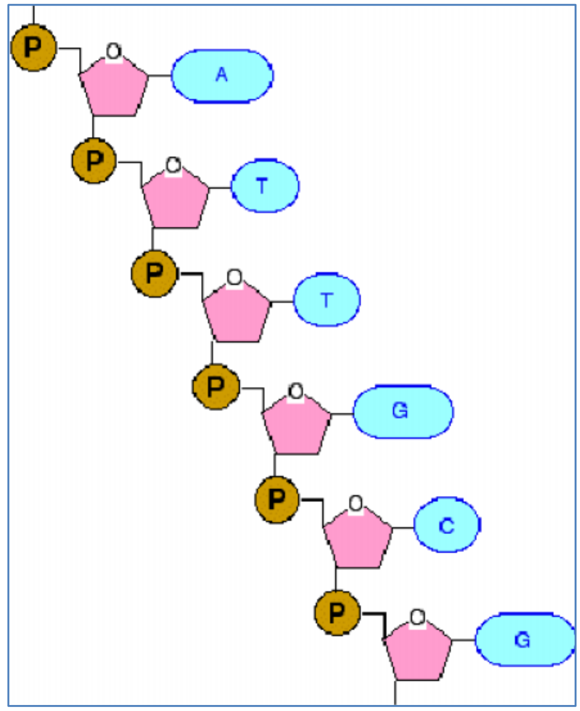
Nucleotides



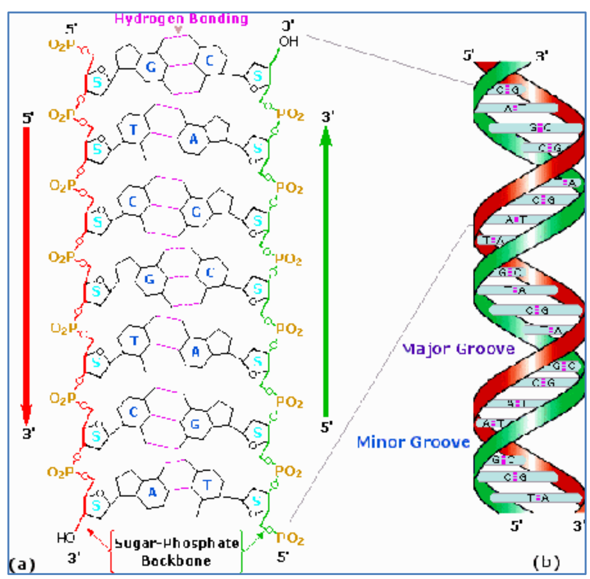
- Nucleic acids are long chain molecules formed of several repeating units called **Nucleotides**.
- The **Nucleotides** are composed of 3 important chemical components:
    - **Sugar**: It is a pentose sugar (5 carbon atoms) having a pentagonal ring structure. It is Deoxyribose sugar in DNA while in RNA it is Ribose sugar.
    - **Phosphate**: It is present as Phosphoric acid (H_3PO_4).
    - **Nitrogen Bases**: Based on their chemical structure they are grouped into 2 classes:
        - Purines which include Adenine (A) and Guanine (G) and
        - Pyrimidines which include Cytosine (C), Thymine (T) and Uracil (U).



- Pyrimidines are monocyclic nitrogenous bases while Purines are bicyclic bases.
- The base pairing of the nitrogen bases are specific. Adenine (A) pairs with Thymine (T) in DNA while in RNA it pairs with Uracil (U) and Cytosine (C) pairs with Guanine (G).
- Nucleotides are joined together by a **Condensation Reaction** between the Phosphate Group of one and the Sugar Group of another.
- The bond between the phosphate from the 5´´ position of one nucleotide and the 3 ´ ´ hydroxyl of the preceding nucleotide called a **Phosphodiester Bond**.
- Many nucleotides joined together in this way make a repeating Sugar-Phosphate 'backbone' out of which the organic bases project.



- The ribose-phosphate backbone has the same shape for both A:T pairs and G:C pairs.
- Consequently, all nucleic acids have a free end composed of the 5’ phosphate group and a 3’ hydroxyl group. The nucleic acid strands have a 5’← -> 3’ orientation.
- Most DNA molecules have much higher numbers of monomer units than are present in any known protein.
- **Nucleosides**: referred to a molecule composed of a sugar and a nitrogenous base having a glycosidic linkage. It includes Adenosine, Guanosine, Thymidine and Cytidine.



Deoxyribonucleic acid (DNA)

- DNA containing the genetic instructions used in the development and functioning of all known living organisms
- The DNA segments carrying this genetic information are called genes.
- DNA consists of two long polymers of simple units called nucleotides, with backbones made of sugars and phosphate groups joined by ester bonds.
- These two strands run in opposite directions to each other and are therefore anti-parallel.
- This information is read using the genetic code, which specifies the sequence of the amino

acids within proteins.

- Within cells DNA is organized into long structures called chromosomes.
- Eukaryotic organisms (animals, plants, fungi, and protists) store most of their DNA inside the cell nucleus and some of their DNA in organelles, such as mitochondria or chloroplasts.
- In contrast, prokaryotes (bacteria and archaea) store their DNA only in the cytoplasm.

Ribonucleic acid (RNA)

- RNA contains the bases A, G, C, and U instead of A, G, C, and T.
- RNA exists as a single strand instead of a double strand.

- There are three primary types of RNA. They are

  (1) messenger RNA (m-RNA)

  (2) ribosomal RNA (r-RNA)

  (3) transfer RNA (t-RNA).

- The different kinds of RNA perform different functions
- **RNA makes protein synthesis possible**: The three RNA make it possible for the encoded information carried by the DNA to be put to use in the synthesis of proteins.
- **Transfer RNA (tRNA**) are the smallest of the three types (73-93 nucleotide residues), and they ***carry*** the correct amino acid to the site of protein synthesis.
- **Messenger RNA (mRNA)** are of variable size, depending on the protein to be manufactured, and contain the information that specifies which protein should be made.This ***message*** is a sequence of RNA nucleotides that is complementary too the template strand of DNA.
- **Ribosomal RNAs (rRNA)** are relatively long RNA strands (hundreds or thousands of nucleotide residues) that combine with proteins to form ribosomes, the multisubunit complexes in which protein synthesis takes place.

Characteristic	**RNA**	**DNA**
Bases	RNA does not contain the base thymine; instead, it substitutes the base uracil	DNA does not contain the base uracil; instead it substitutes the base thymine
Strands of Nucleotides	RNA is *typically* single stranded	DNA is *typically* double stranded
Oxygens	RNA has one more oxygen in the ribose than DNA	DNA has one less oxygen in the ribose than RNA

7. VITAMINS AND MINERALS

Vitamins

- Vitamins and minerals are substances your body needs in small amounts for normal growth, function and health. Together, vitamins and minerals are called micronutrients.
- Both vitamins and minerals are considered coenzymes because they assist the enzymes in the body in carrying out all of the daily activities.
- They are needed to help convert carbohydrates, fat and protein into energy.
- Your body can't make most micronutrients, so you must get them from the foods you eat or, in some cases, from supplements. They are a vital part of a healthy diet.
- Research shows that a diet rich in vitamins and minerals is linked with prevention of certain diseases such as heart disease and cancer.
- Whole foods like fruit, vegetables and whole grains provide a complex combination of vitamins, minerals, fiber and other substances that promote health.
- By consuming a variety of foods from all the food groups, most people can achieve their vitamin and mineral needs. However, if you feel you are unable to reach your vitamin and mineral needs through diet, you may need a supplement.
- It is important to remember that supplements are used to compliment your diet, not replace it.
- Avoid taking overdoses of vitamins and minerals and be sure to read the dosage on the label. The best way to compliment your diet is a multivitamin.

Classification

Vitamins are essential micronutrients, required by the body in small amounts.Vitamins are broken into two categories: Water-soluble and fat-soluble.

Water-soluble vitamins

- Vitamin C , biotin and the seven B-complex vitamins--thiamin (B-1), riboflavin (B-2), niacin (B-3), pantothenic acid (B-5), pyridoxine (B-6), folic acid (B-9) and cobalamin.
- Dissolve in water and what is not used by the body is simply excreted in your urine.

Fat-soluble vitamins

- Vitamins A, D, E, and K
- Can accumulate and may become toxic in high doses. For example, vitamins E and K affect blood clotting and therefore can interact with certain medications such as Coumadin, which is used as a blood thinner.

Vitamin	Major Source	Benefits	Deficiency
Vitamin A	Liver, milk, cheese, Plant foods that are orange, red and deep green	Essential for night vision, bone development, maintains healthy tissue	Night blindness , Dryness of eye , Risk of infections, Poor growth in children
Vitamin D	fortified milk and cereals, fatty fish like salmon, tuna, halibut, fish oil, egg yolk, sunlight	Increases absorption of Calcium, bone and tooth health ,	Abnormal bone growth & repair , Rickets in children ,Osteomalacia in adults , Muscle spasms.
Vitamin E (tocopherol)	Vegetable oils (nuts, almonds, sunflower etc.), margarine, green leafy vegetables (spinach etc), wheat germ, whole grain products	Antioxidant, Fighting toxins, maintaining healthy nerves, skin, muscles, red blood cells	Weak muscles , Fertility problems , Slow healing of wounds
Vitamin K	Liver, soy bean oil, spinach, cabbage, asparagus, green leafy vegetables	Aids in blood clotting	Bleeding
Vitamin C (Thiamine)	Citrus fruits, tomatoes, potatoes, cabbage, green peppers, broccoli, strawberries	Antioxidant, wound healing, aids in iron absorption , Bone & connective tissue growth, Function of blood vessels.	Scurvy (bleeding, loose teeth, & inflammation of the gums)
Vitamin B1 (Thiamine)	Dried yeast, whole grains, meat, nuts, legumes, & potato.	Carbohydrate metabolism , Nerve & heart function	Beriberi disease, irritation, poor memory, sleep disturbances, pain in the chest
Vitamin B2 (Riboflavin)	Milk, cheese, eggs, fish, liver, enriched cereals, brewer's yeast, almonds, meat, mushrooms	Healthy skin, aids in energy production in cells of our bodies (Carbohydrate metabolism), Healthy hair and nails ,	Photophobia, excess tears, itching of the eyes, burning of the lips and tongue.
Vitamin B3 (Niacin or Nicotinic Acid)	Dried yeast, liver, meat, fish, legumes, and whole-grain enriched cereal products.	Could help aid in heart disease by blocking free fatty acids(cholesterol), helps to maintain healthy intestinal tract and healthy skin.	Pellagra (dermatosis, inflammation of the tongue, abnormal intestinal and brain function) .
Vitamin B6 (Pyridoxine)	Dried yeast, liver, organ meats, wholegrain cereals, fish, & legumes.	Protein metabolism, hemoglobin and red blood cell formation, Healthy Nervous system , Healthy skin	Anemia, sleeplessness, irritability, anxiety, dry skin, cracked lips, stress and premenstrual symptoms are all possible symptoms.
Vitamin B12 (Cobalamin)	Liver, beef, pork, poultry, dairy products, eggs	Maturation of red blood cells , Nerve function , DNA synthesis, healthy blood.	Anemia , Some psychiatric disorders , Poor vision
Biotin	Liver, kidneys, egg yolks, yeast, nuts, and legumes	Carbohydrate and fatty acid metabolism	Inflammation of the skin and lips
Folic acid	Fresh green leafy vegetables, fruits, liver, and dried yeast , broccoli, pastas and cereals	Maturation of red blood cells, DNA and RNA synthesis, Prevents birth defects, aids in treatment of anemia	Decrease in the number of all types of blood cells , Large red blood cells
Pantothenic acid	Liver, yeast, and vegetables	Carbohydrate and fat metabolism	Neurologic disease , Burning feet

Minerals

- Minerals are the main components in your teeth and bones, and they serve as building blocks for other cells and enzymes.
- Minerals also help regulate the balance of fluids in your body and control the movement of nerve impulses.
- Some minerals also help deliver oxygen to cells and help carry away carbon dioxide.
- Deficiencies of minerals, except iron & iodine, are uncommon.
- Excesses of some minerals may cause toxicity.

Minerals have two categories:

- Major minerals (macronutrients) and
- trace minerals(micronutrient) .

Major minerals

- also called macronutrients
- Calcium, phosphorus, magnesium, sodium, potassium, sulfur and chloride are considered major minerals because adults need them in relatively larger amounts - more than 250 milligrams (mg) a day.

Trace minerals

- also called micronutrient
- Chromium, copper, fluoride, iodine, iron, manganese, molybdenum, selenium and zinc are considered trace minerals because your body needs them in smaller amounts -- fewer than 20 mg a day.

Mineral	Major Source	Benefits	Deficiency
Sodium	Salt, beef meat, sardines, cheese, green olives, & corn bread	Acid-base balance , Nerve & muscle function ,	Low sodium levels in the blood , Confusion , Coma
Potassium	Milk, bananas, prunes, and raisins	Nerve and muscle function , Acid-base & water balance	Low potassium levels in the blood , Paralysis , Heart disturbances
Calcium	Milk & milk products, meat, fish, egg, cereal products, beans, fruits, & vegetables , dried peas and almonds.	Healthy bone & tooth formation , Blood clotting , Nerve & muscle function , Normal heart rhythm	Low calcium levels in the blood , Muscle spasm
Phosphorus	Protein foods: meat, poultry, fish, eggs, milk, cheese, butter, nuts, ice cream, chocolate, peanut	Bone & tooth formation , Acid-base balance , Component of nucleic acids , Energy production	Irritability , Weakness , Blood cell disorders , Abnormalities of the intestine & kidney
Iron	Soybean flour, Organ meats, beef, poultry, shellfish, dried peas and beans, whole-grain products, green leafy vegetables	Oxygen transport , Formation of enzymes , The main component of red blood cells and muscle cell	Anemia , Difficulty in swallowing , Spoon-shaped nails , Decreased work performance , Impaired learning ability
Magnesium	Avocado, bananas, Milk, yogurt, dried beans (garbanzo beans, lentils, kidney beans), nuts (pumpkin seeds, almonds, cashews, flaxseed), whole grains, tofu, spinach, potato with skin	Protein Synthesis, production and transportation of energy , Bone & tooth formation , Nerve & muscle function , Enzyme activation	Low magnesium levels in the blood , Abnormal nerve function
Zinc	Organ meats, and seafood ,meat, fish, poultry, shellfish, nuts, whole-grain precuts	Energy metabolism, immune function, wound healing, taste and smell , component of enzymes and insulin , Healthy skin	Slowed growth , Delayed sexual maturation , Diminished taste sensation
Copper	Organ meats, nuts, dried legumes, and whole-grain cereals.	Enzyme component , Formation of red blood cells , Bone formation	Anemia in malnourished children
Manganese	Whole-grain cereals, and dried fruits	Enzyme component	Weight loss , Skin irritation , Nausea & vomiting , Changes in hair color, slow hair growth.
Selenium	Meats and other animal products	Necessary for synthesis of an antioxidant enzyme	Muscle pain & weakness
Iodine	Seafood, iodized salt, & dairy products	Formation of thyroid hormones	Enlargement of the thyroid gland , Cretinism , Deaf-mutism , Abnormal fetal growth & brain development
Fluoride	Milk, egg yolks, drinking water, seafood , Tea, coffee, & fluoridated water	Bone & tooth formation ,	Increased risk of dental cavities , Possibly bone thinning